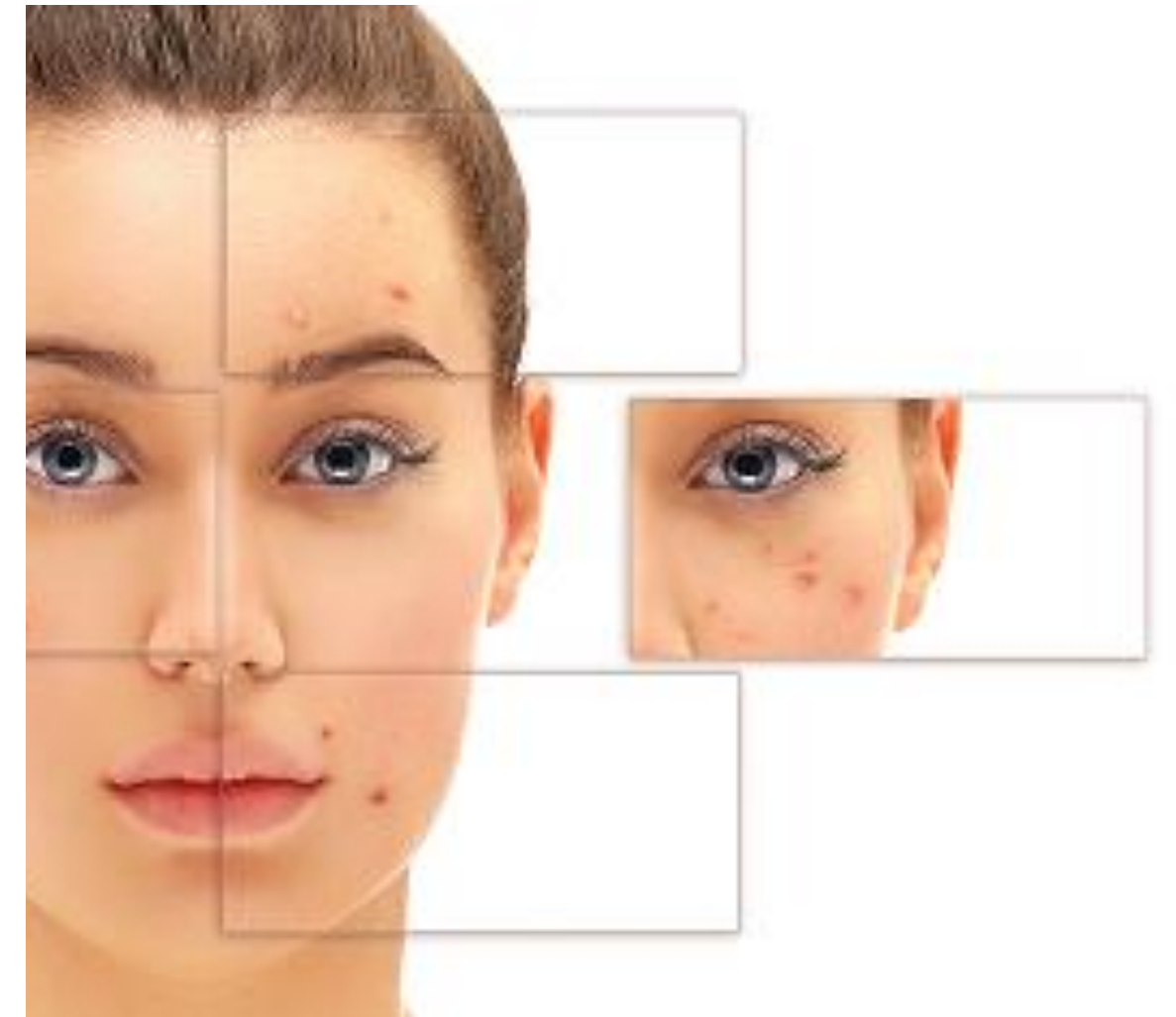


Multimodal AI Framework for Personalized and Culturally Aware Skin Disease Diagnosis, Monitoring, and Treatment Support.



GROUP : 25-26J-222



GROUP MEMEBERS



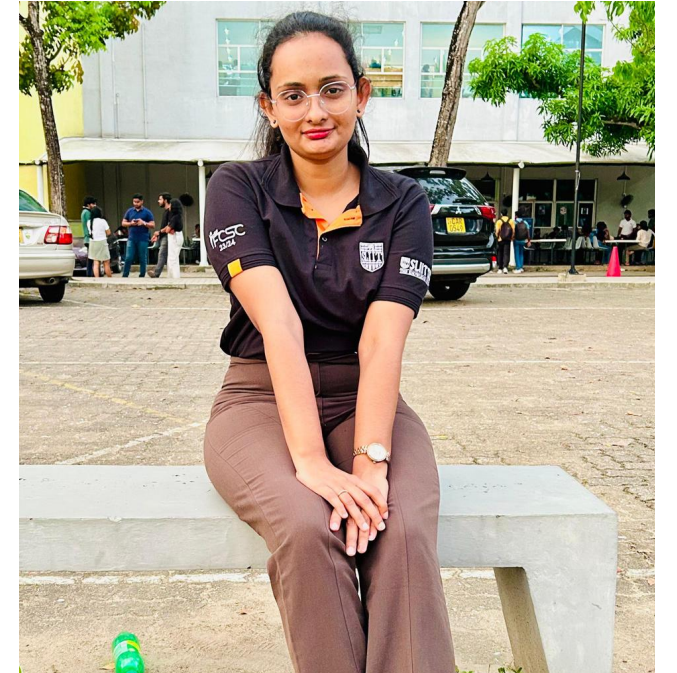
Mihisandali W.K.M



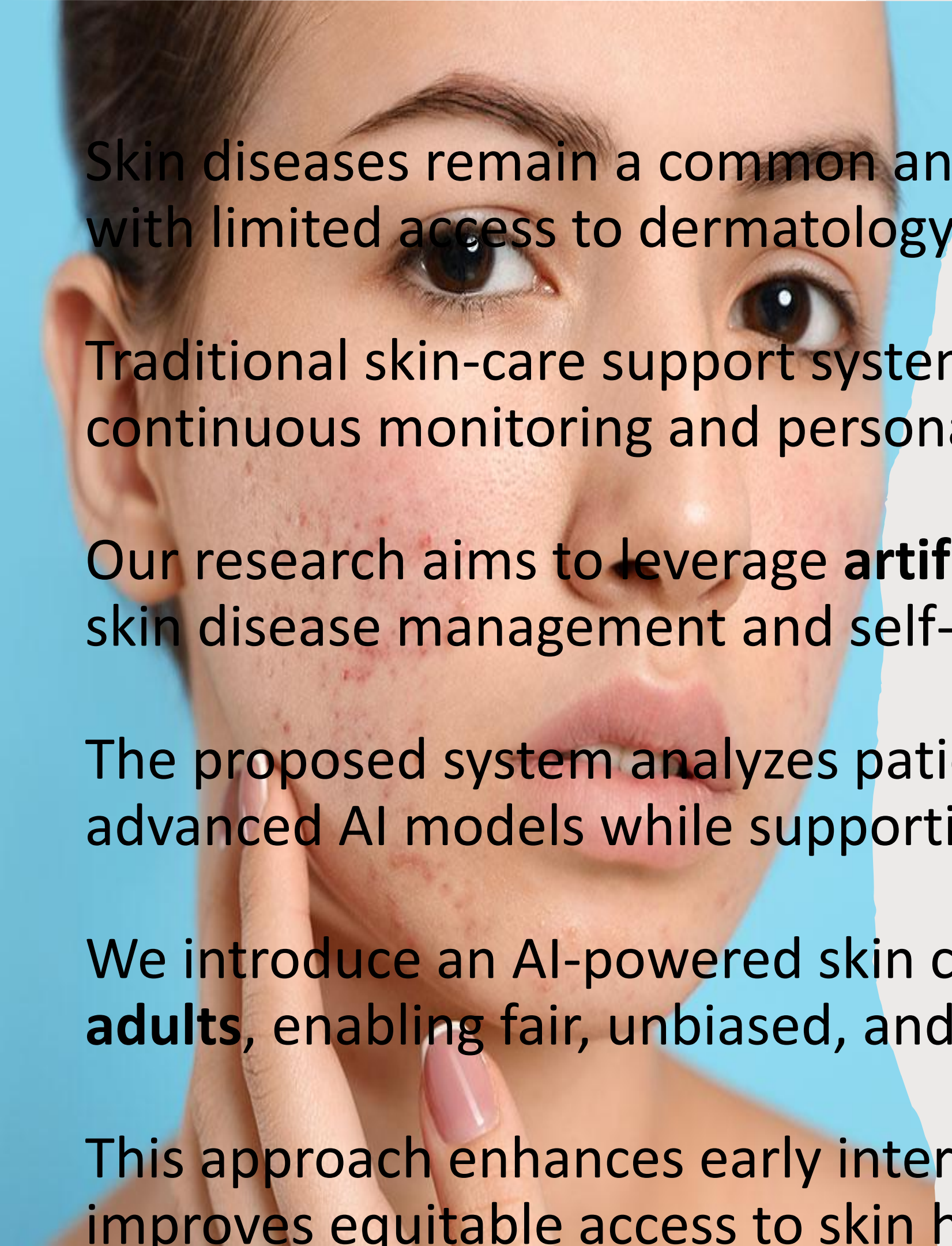
Laksopan R



Wijesinghe
H.W.M.O.P



Devindi K.T.P



INTRODUCTION

Skin diseases remain a common and growing health concern, especially in regions with limited access to dermatology specialists.

Traditional skin-care support systems focus mainly on diagnosis and often lack continuous monitoring and personalized treatment guidance.

Our research aims to leverage **artificial intelligence and machine learning** to improve skin disease management and self-care support.

The proposed system analyzes patient symptoms and optional skin images using advanced AI models while supporting multilingual interaction.

We introduce an AI-powered skin care support platform designed for **teenagers and adults**, enabling fair, unbiased, and continuous monitoring.

This approach enhances early intervention, reduces unsafe self-medication, and improves equitable access to skin health support.

RESEARCH PROBLEM

Many people suffer from common skin diseases but lack timely access to dermatology specialists, especially in rural and underserved areas.

Traditional skin-care systems focus mainly on diagnosis and lack personalized treatment and continuous monitoring, leading to ineffective self-care.

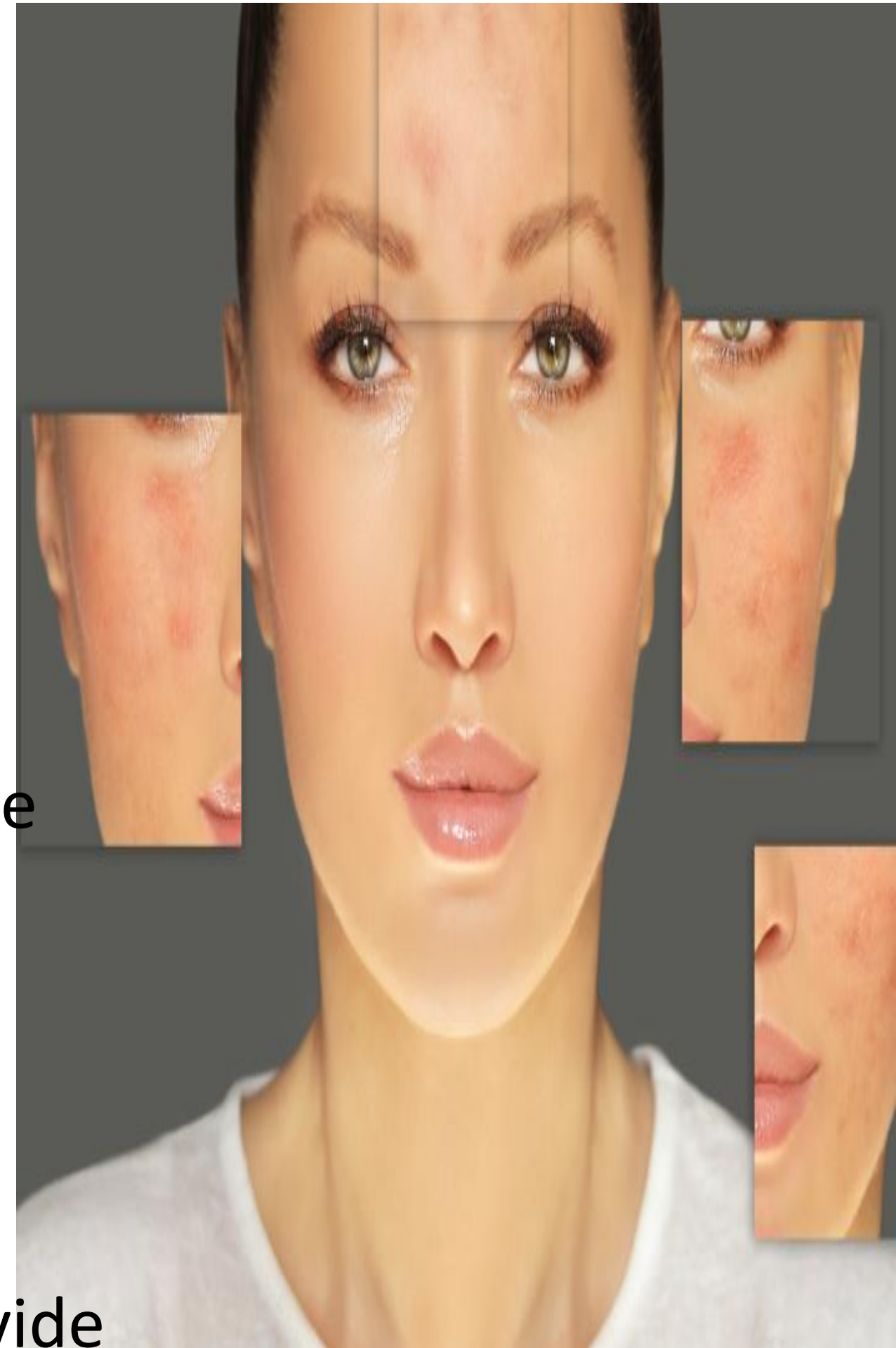
Existing AI-based dermatology solutions often perform poorly across different skin tones, resulting in biased and unreliable outcomes for diverse populations.

Language barriers and lack of cultural adaptation reduce the usability of current digital health systems for local users.

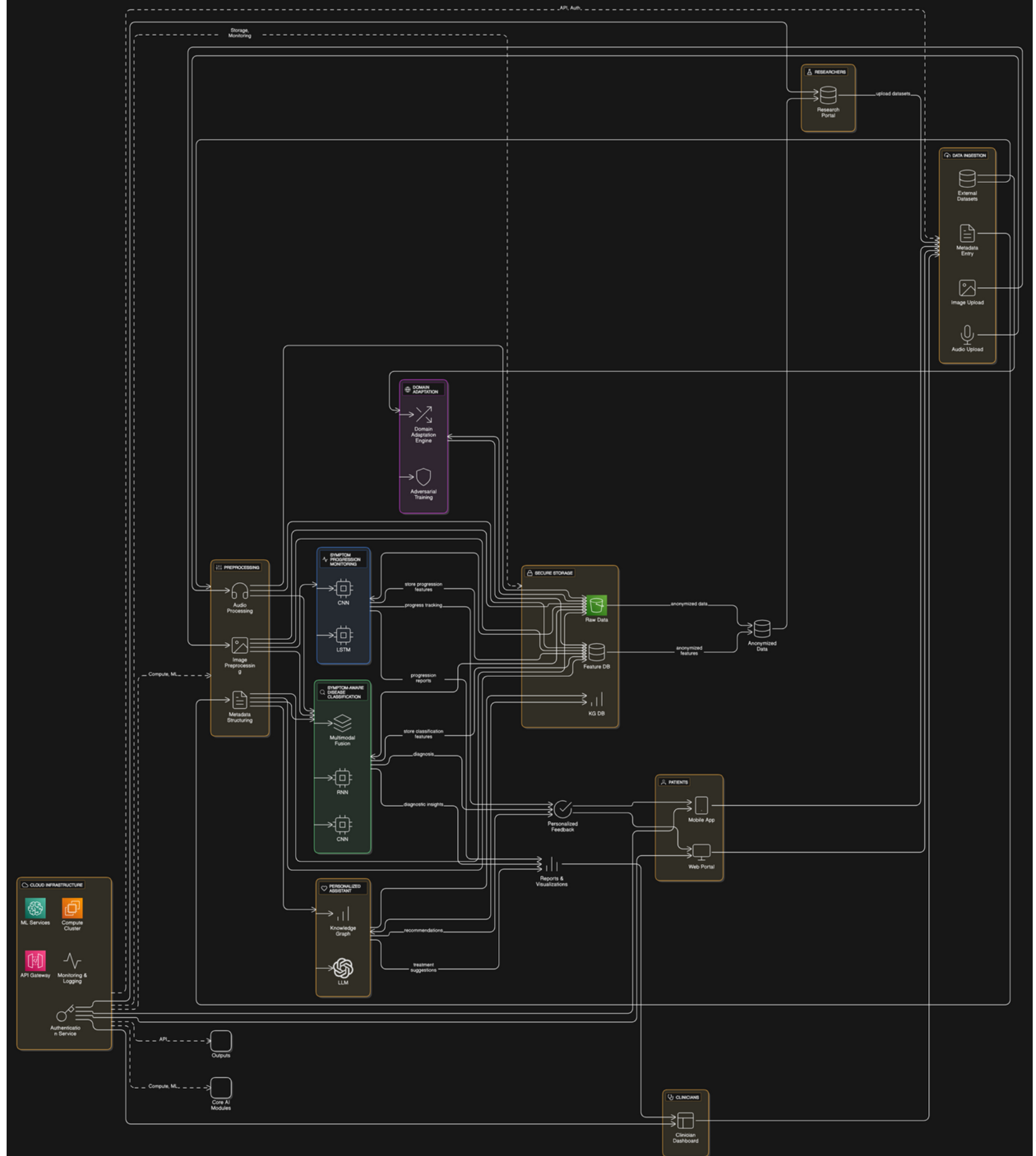
The absence of continuous monitoring mechanisms makes it difficult to track disease progression, causing delayed intervention and worsening of skin conditions.

OBJECTIVES

- Implement machine learning and deep learning models to analyze **skin images and symptom descriptions** accurately.
- Enable **multilingual support (Sinhala and English)** to improve accessibility for local users.
- Reduce **skin-tone bias** and ensure fair model performance across diverse populations.
- Implement continuous monitoring to track skin condition progression over time.
- Develop an **AI-powered skin care support system** to provide safe and personalized treatment guidance.



SYSTEM OVERALL DIAGRAM





IT 22246400

Mihisandali W.K.M

B.Sc. (Hons) Degree in Information Technology Specializing in Information Technology

Personalized Treatment Support & Monitoring

INTRODUCTION



AI-powered skin care support system for Sri Lankan patients

Provides safe, personalized, and culturally adapted treatment guidance

Supports self-care advice, safe medicine/OTC suggestions, and progress monitoring

Multilingual system with **Sinhala and English** support

Unlike diagnosis-only tools, offers **treatment-focused and lifetime patient support**

Problems

Most existing AI-based skin care systems **focus only on diagnosis**, not treatment support.

No follow-up mechanism after initial guidance is provided.

Patients do **not receive**:

- Ongoing treatment support

- Progress monitoring

- Safety reminders or escalation alerts

Sub-Problem / Prototype Focus

Absence of **continuous treatment monitoring**

Lack of **patient engagement after first recommendation**

No system to detect **worsening conditions or red-flag symptoms**



Proposed Solution:

✓ AI-Driven Treatment & Follow-Up Support System

Provide **safe self-care advice and OTC medicine suggestions** tailored to the patient.

Introduce **notification-based follow-ups** to monitor progress:

- Day 1** – Check initial treatment response

- Day 3** – Identify improvement or worsening

- Day 7** – Decide recovery or recommend medical escalation.

OBJECTIVES

Main objective:

To develop an AI-powered system that provides **safe, personalized, and culturally adapted treatment guidance** for skin diseases in Sri Lanka.



Sub Objectives:

Personalized Treatment Guidance.
Cultural and Language Adaptation.
Continuous Patient Support.
User Safety and Reliability.
Improving Access to Care



Technologies Use in the system

1. Current Technologies

Machine Learning / NLP:

TF-IDF (Term Frequency–Inverse Document Frequency) → Converts Sinhala/English text input into numerical features for model processing.

Logistic Regression → Predicts personalized treatment suggestions based on features.

Calibrated SVM (Support Vector Machine) → Improves prediction confidence and ensures safer recommendations.

Backend / Frontend Development:

VS Code → Development environment for coding and project management.

Backend (Python / Flask or FastAPI) → Handles API calls, data processing, and model inference.

Frontend (Flutter) → Builds mobile application.

Version Control & Collaboration

Git

2. Future Technologies

Machine Learning / NLP:

Multilingual Transformer Models (mBERT, XLM-R, IndicBERT) → Sinhala/English text understanding for improved accuracy.

Lightweight CNN / Vision Transformers (ViT) → Optional skin image classification for multi-modal support.

Rule-Augmented NLP Pipelines → Red-flag detection and escalation triggers for patient safety.

Knowledge Graph + LLM Integration → Personalized treatment suggestions with cultural adaptation.

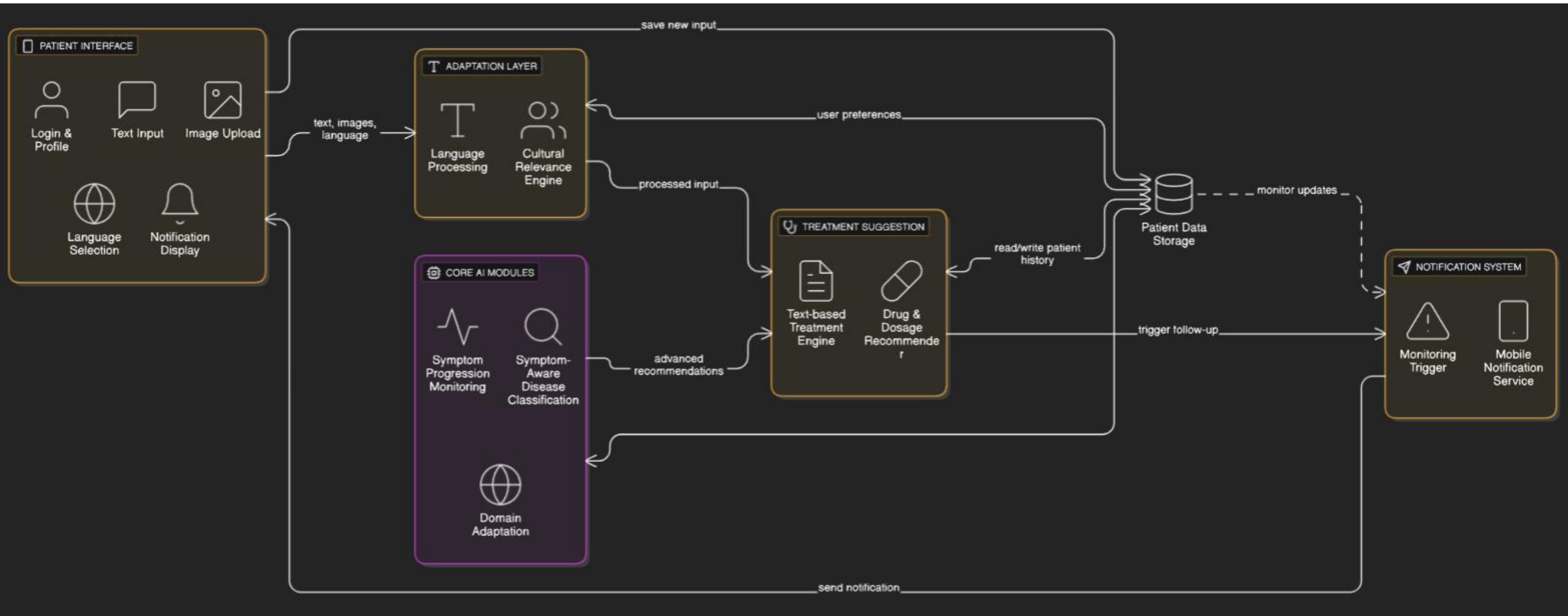
Backend / Frontend / Deployment:

Notification APIs & Mobile Frameworks → Continuous patient monitoring (Day 1/3/7 follow-ups).

REST APIs (Planned) → Will connect the AI model to the frontend for real-time suggestions.

Implementation of the project

- SYSTEM OVERVIEW DIAGRAM



Progress Overview

```
01_preprocessing.ipynb
File Edit View Insert Runtime Tools Help

Q Commands + Code + Text ▶ Run all

[ ] ▶
parts = []
preferred_cols = [
    "age", "gender", "sex", "body_part", "location", "site", "duration",
    "itching", "pain", "burning", "fever",
    "rash_type", "lesion_type", "appearance",
    "symptoms", "symptom_text",
    "history", "notes", "description", "case_text"
]
for c in preferred_cols:
    if c in row.index and pd.notna(row[c]) and str(row[c]).strip():
        parts.append(f"{c.replace('_', ' ')}: {row[c]}")

# fallback: include other non-empty fields (except label-like)
if len(parts) < 2:
    for c in row.index:
        if c.lower() in ["label", "disease", "condition", "diagnosis", "dx", "final_diagnosis"]:
            continue
        v = row[c]
        if pd.notna(v) and str(v).strip():
            parts.append(f"{c}: {v}")

return " | ".join(parts).strip()

def detect_label_column(columns):
    """
    Robust label/diagnosis column detection.
    """
    cols = list(columns)
```

preprocessing

Treatment suggesion

```
03_treatment_recommendation.ipynb
File Edit View Insert Runtime Tools Help

Q Commands + Code + Text | ▶ Run all

- Impetigo
- Lichen planus
- Molluscum contagiosum
- Nummular eczema
- Perioral dermatitis
- Photodermatitis
- Pityriasis versicolor
- Psoriasis vulgaris
- Rosacea

=== Treatment Suggestions for: Acne vulgaris (severity=moderate) ===
Age : 22
Allergies: ['benzoyl peroxide']
Prefer OTC: True

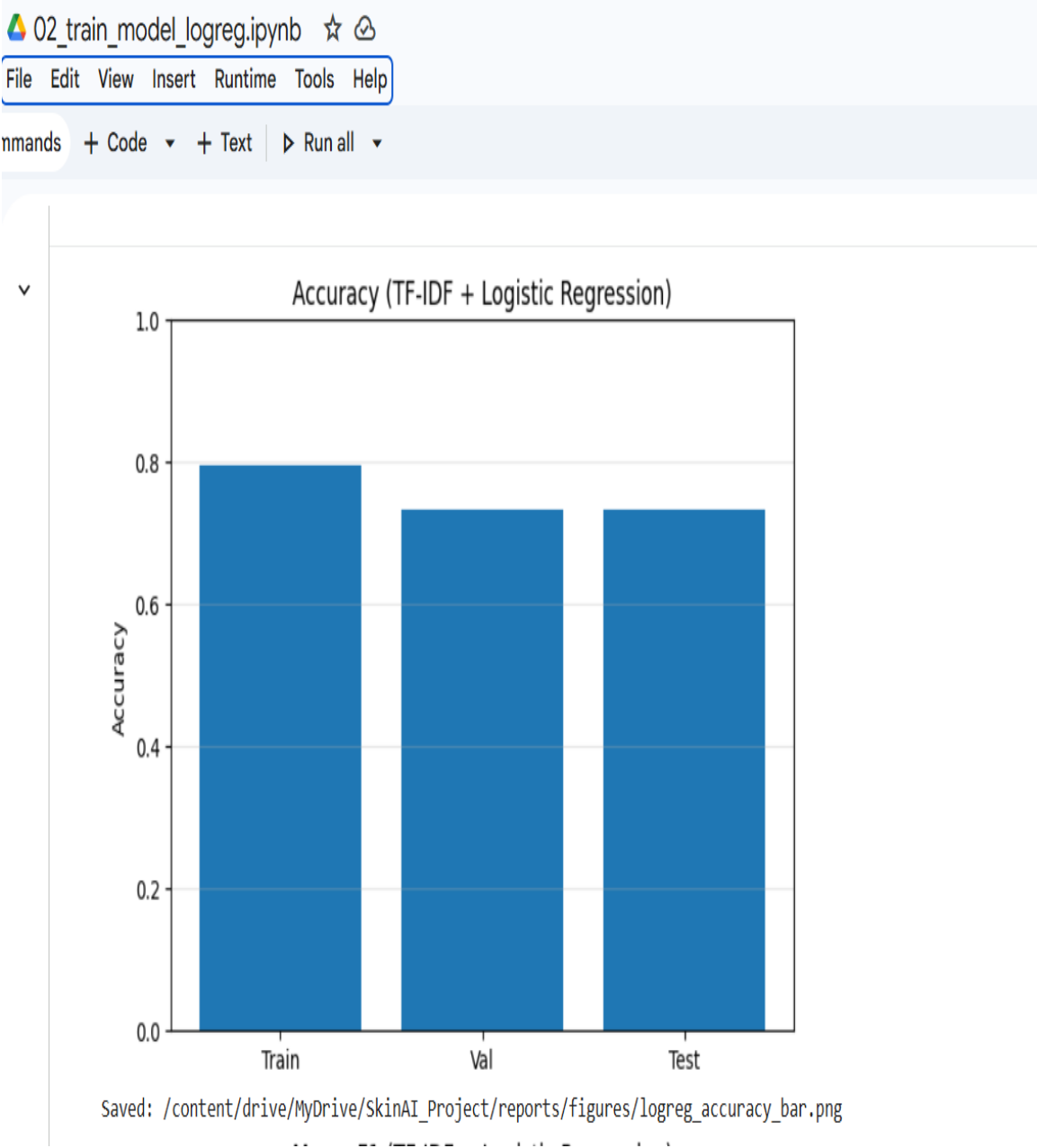
- topical retinoid | severity=moderate | OTC=yes | support_count=11
- salicylic acid wash | severity=moderate | OTC=yes | support_count=10

=== Treatment Suggestions for: Atopic dermatitis (Eczema) (severity=mild) ===
Age : 5
Allergies: []
Prefer OTC: False

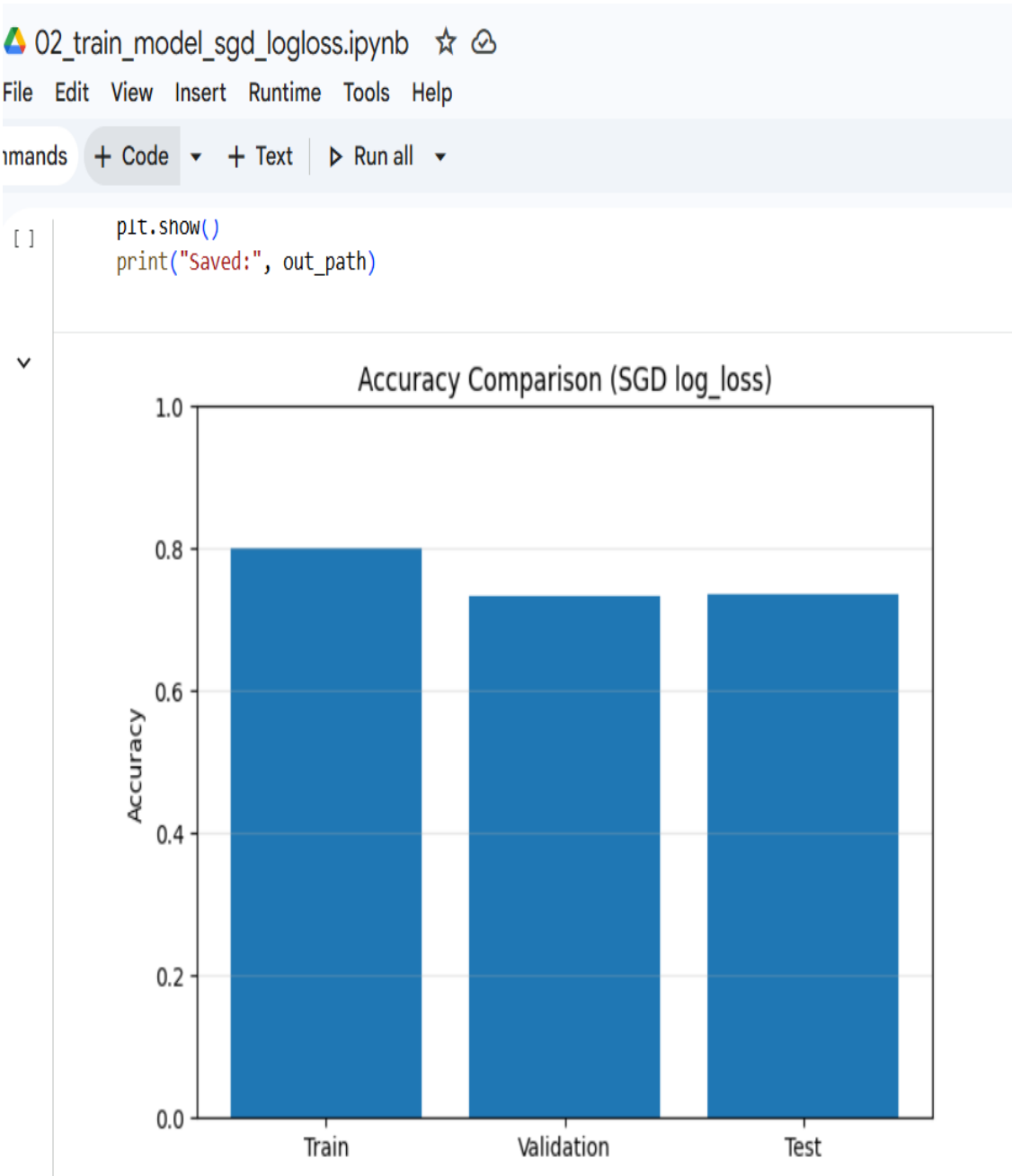
- emollient cream | severity=mild | OTC=yes | support_count=23
- topical steroid (low potency) | severity=mild | OTC=yes | support_count=18
- fragrance-free moisturizer | severity=mild | OTC=yes | support_count=15

grouped_path = os.path.join(SYNTHETIC_DIR, "treatment_stats_grouped.csv")
grouped.to_csv(grouped_path, index=False)
print("Saved treatment stats to:", grouped_path)
```

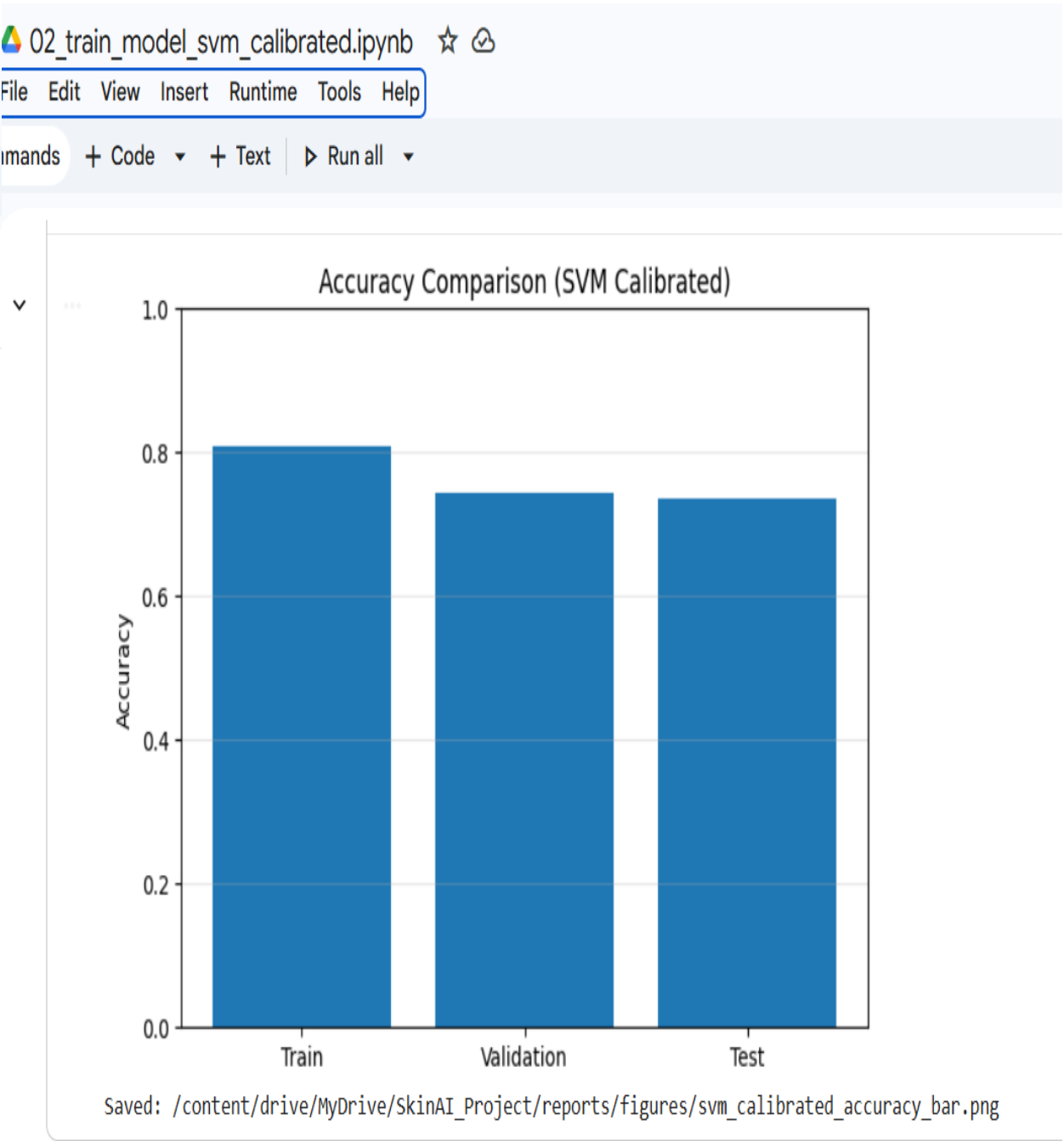
Progress Overview



Model 1



Model 2



Model 3

Progress Overview

```
06_kb_ga_chatbot.ipynb ☆
File Edit View Insert Runtime Tools Help

nmands + Code + Text ▶ Run all

[ ] kb_meds = pd.read_csv(os.path.join(KB_DIR, "kb_medicines_10000.csv"))
    kb_ings = pd.read_csv(os.path.join(KB_DIR, "kb_ingredients_10000.csv"))

    print("kb_diseases:", kb_diseases.shape)
    print("kb_meds      :", kb_meds.shape)
    print("kb_ings      :", kb_ings.shape)

    kb_diseases.head(2)

kb_diseases: (10000, 11)
kb_meds      : (10000, 11)
kb_ings      : (10000, 8)

    disease severity skin_type overview causes common_symptoms contagious_info self_care do_and_dont when_to_see_doctor fi
0 Impetigo mild sensitive A contagious bacterial skin infection, often w... Bacteria entering through cuts/scratches; spre... red sores, blisters, honey-colored crusts, itc... Contagious. Avoid sharing towels and close con... Keep area clean, avoid touching, keep nails sh... Do: hygiene; Don't: pick crusts; try to avoid ... See a doctor soon; impetigo often needs prescr... impetigo;mild;s
1 Hives (Urticaria) severe sensitive Raised itchy welts that can appear and fade ov... Allergies, infections, heat, stress, foods, me... itchy raised welts, redness; may move around t... Not contagious. Cool compress, avoid triggers, gentle skincare... Do: try to avoid known triggers; Don't: overhe... Urgent help if swelling of lips/face, breathin... hives;sensitive
```

Functional Requirements

- Accept Sinhala/English text and optional images from patients.
- Provide treatment suggestions → self-care steps, safe OTC options, escalation rules.
- Send follow-up notifications (Day 1/3/7) and track progress.
- Maintain lifetime patient records with consent and disclaimers.

Non-Functional Requirements

- Safety & Accuracy → Red-flag detection $\geq 95\%$, treatment suggestion accuracy $\geq 85\%$.
- Performance & Reliability → Fast response ($< 3s$) and uptime $\geq 99.5\%$.
- Privacy & Security → Data encryption, access control, audit logs.
- Cultural & Language Adaptation → Clear Sinhala/English output, Sri Lanka-specific advice.

Software Requirements

- **Backend** – Python (FastAPI) / Node.js (Express) for APIs.
- **Frontend** – Web interface (React).
- **Database** – MySQL for patient data & treatment records.
- **Storage** – Cloud (for images) with encryption.
- **AI Models** – Multilingual NLP (Sinhala + English), lightweight CNN/ViT for images.
- **Notification Service** – Firebase Cloud Messaging for reminders (Day 1/3/7).
- **Security** – Data encryption, consent handling, access control, audit logs.

Personnel (Human Resource) Requirements

- **AI/ML Engineer** – Builds and fine-tunes NLP & image analysis models.
- **Backend Developer** – Implements APIs, database, and business logic.
- **Frontend Developer** – Creates patient-facing chatbot UI and notifications.
- **Database Administrator (DBA)** – Ensures secure, reliable patient record storage.

Real-World problem

Scenario

Patients develop skin conditions (rash, eczema, acne) and seek guidance online or via AI apps.
Most existing AI systems **only provide one-time diagnosis**, without treatment guidance.
No follow-up, progress monitoring, or safety alerts for worsening symptoms.
Patients may **misuse medicines, stop treatment prematurely, or delay seeing a doctor**.

Challenges

Multilingual Support: Accurate understanding of **Sinhala and English** input.

Safety & Reliability: Ensuring treatment advice is **medically safe** and **culturally appropriate**.

Patient Engagement: Encouraging adherence to notifications and treatment recommendations.

Real-World Impact

Improved Treatment Adherence: Continuous guidance ensures correct use of self-care and medications.

Early Detection of Complications: Red-flag alerts prompt timely medical intervention.

Cultural & Language Adaptation: Sinhala and English guidance increases trust and usability.

Commercialization

Investment & Cost Recovery

- **Investment/Cost:**
 - *Labor:* AI/ML engineers, developers, domain experts.
 - *Material:* Cloud hosting, storage, APIs, security & compliance.
- **Cost Recovery:**
 - **Freemium model** – basic advice free, premium features paid.
 - **Subscriptions** – low-cost monthly/annual plans.

Target Market:

- Urban & rural Sri Lankan population with skin care needs.
- Public health sector & NGOs promoting affordable digital health.
- Pharmacies, insurers wanting to add digital tools.

Customer Profile:

- Patients seeking affordable first-level advice.
- Parents managing children's skin conditions.
- Healthcare providers needing a safe triage tool.

Value Proposition:

- Affordable, accessible, safe skin treatment guidance.
- Multilingual + culturally adapted for Sri Lanka.
- Reduces unnecessary doctor visits while still ensuring urgent cases get escalated.

Intellectual Property Rights (IPR):

- Trademark the brand/app name.
- Copyright the chatbot content, templates, and UI design.
- Patent (if possible) for the unique rule-based + multilingual treatment suggestion pipeline.

REFERENCES

- [1] J. Zhou, X. He, L. Sun, J. Xu, X. Chen, Y. Chu, ... X. Gao, "SkinGPT-4: An interactive dermatology diagnostic system with visual large language model," *arXiv preprint arXiv:2304.10691*, 2023.
- [2] N. T. Huynh, *et al.*, "DermAI: A chatbot assistant for skin lesion diagnosis using vision and language models," in *Proc. Asian Conf. Comput. Vis. (ACCV) Workshops*, 2024.
- [3] S. P. Choy, *et al.*, "Systematic review of deep learning image analyses for the diagnosis and monitoring of skin disease," *npj Digital Medicine*, vol. 6, no. 1, p. 7, 2023.
- [4] H. Ujiie, *et al.*, "Unmet medical needs in chronic, non-communicable inflammatory skin diseases," *Frontiers in Medicine*, vol. 9, p. 875492, 2022.
- [5] M. A. Kassem, K. M. Hosny, R. Damaševičius, and M. M. Eltoukhy, "Machine learning and deep learning methods for skin lesion classification and diagnosis: A systematic review," *Diagnostics*, vol. 11, no. 8, p. 1390, 2021.
- [6] A. Sarabadani, M. A. Shamami, H. Sadeghsalehi, B. Asadi, and S. Hesarakhi, "DKG-LLM: A framework for medical diagnosis and personalized treatment recommendations via dynamic knowledge graph and large language model integration," *arXiv preprint arXiv:2508.06186*, 2025.
- [7] Mayo Clinic Platform, "Knowledge graphs can move healthcare into the future," *Mayo Clinic Platform Blog*, Dec. 2023.

Assistant Chat

Hello! How can I help you with your skin routine today?

hi

I'm a dummy AI. I recommend seeing a dermatologist for accurate advice.

i have iching

I'm a dummy AI. I recommend seeing a dermatologist for accurate advice.

Ask a question...

 New Scan

 Home

 Analyze

 Chat



IT22277954
LAKSOPAN R

B.Sc. (Hons) Degree in Information Technology
Specializing in Information Technology

INTRODUCTION

- **From Snapshots to Sequences**

Moves beyond single static images to track skin conditions over time.

- **Temporal Awareness**

Analyzes sequential images to detect subtle changes in symptoms day by day or week by week.

- **Progress Tracking**

Monitors improvement, stability, or worsening of skin conditions using measurable parameters.

- **Data-Driven Insights**

Quantifies changes in lesion size, color, or texture to understand disease progression.

- **Inclusive Monitoring**

- Ensures reliable tracking across diverse skin tones for fair assessment.



Early Detection & Treatment Monitoring via Sequential Image Analysis

Problem

1. Limited to Single Images

1. Existing dermatology AI only analyzes one snapshot, missing how skin conditions evolve over time.

2. No Progress Tracking

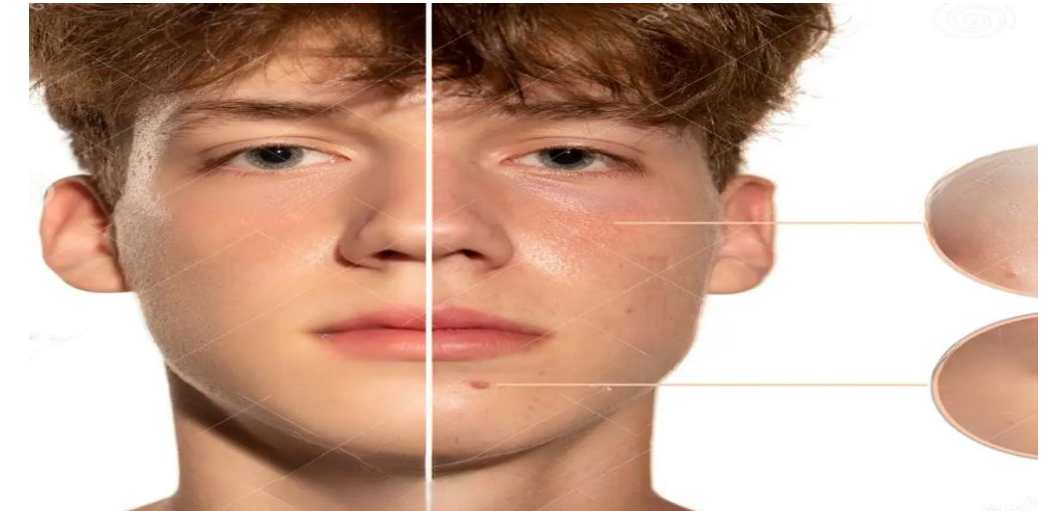
1. Patients and doctors cannot see whether symptoms are improving, stable, or worsening.

3. Subtle Changes Are Missed

1. Small changes in lesion size, color, or texture may go unnoticed without sequential analysis.

4. Bias Across Skin Tones

1. Systems may fail to accurately detect progression in darker or underrepresented skin tones.



Solution

1. Sequential Image Analysis

1. Track skin condition changes over time using daily or weekly image sequences.

2. Temporal Awareness

1. AI identifies patterns of progression, improvement, or deterioration, providing objective monitoring.

3. Parameter-Based Evaluation

1. Measures lesion size, color, and texture to quantify changes accurately.

4. Inclusive Monitoring

1. Ensures consistent and fair tracking across all skin tones, improving reliability

OBJECTIVES

Main Objective

To develop an AI-based sequential image monitoring system for tracking skin disease progression over time.



Sub-Objectives

Temporal Analysis

Analyze sequential images to detect subtle changes in skin conditions day by day or week by week.

Progress Quantification

Measure parameters like lesion size, color intensity, and texture to evaluate improvement, stability, or worsening.

Pattern Recognition

Identify trends and patterns in symptom progression for better understanding of disease dynamics.

Fair Monitoring Across Skin Tones

Ensure accurate tracking for all skin types, reducing bias and improving inclusivity.

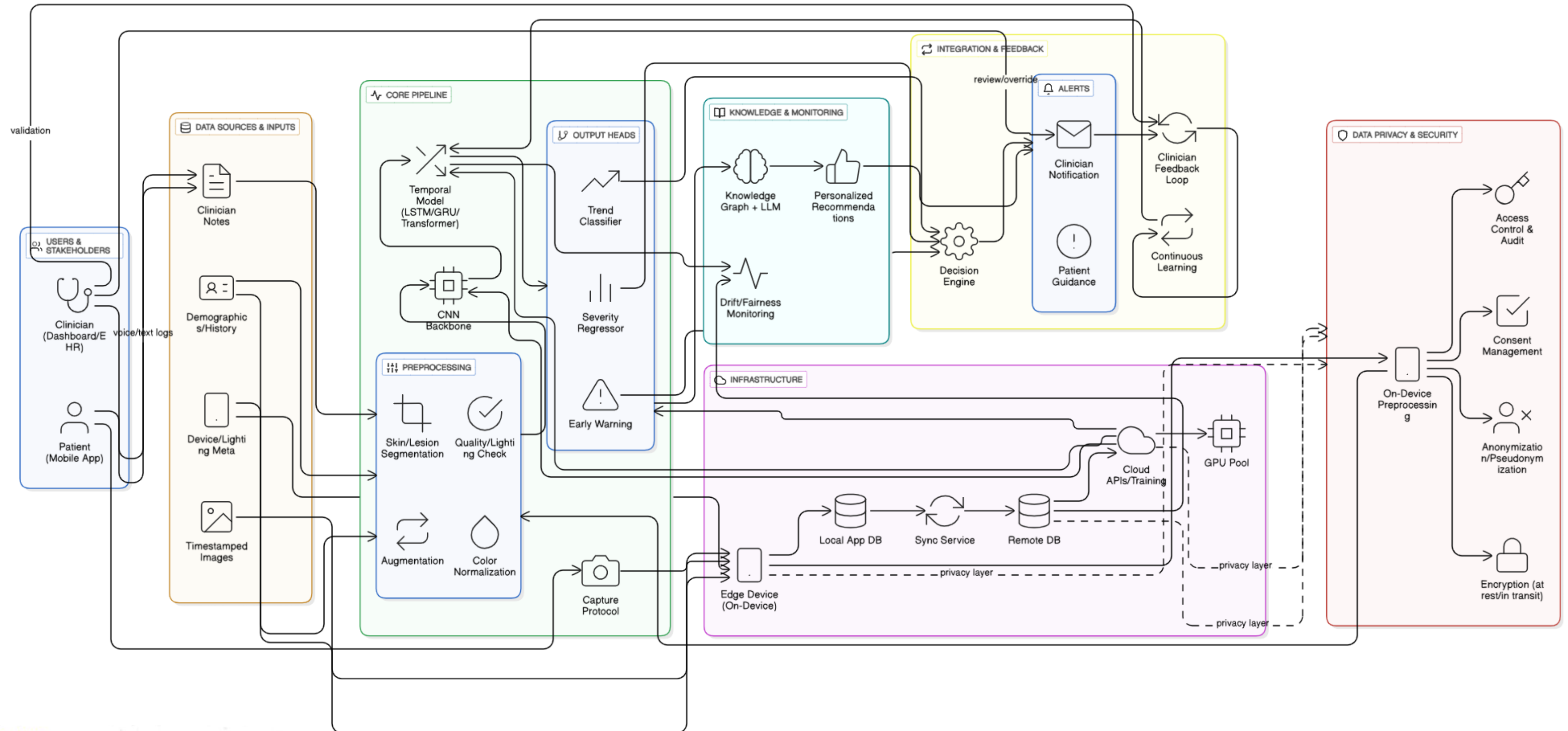
Data-Driven Insights for Monitoring

Generate objective, measurable reports that visually represent progression over time.

Technologies Use in the system

Early Detection & Treatment Monitoring via Sequential Image Analysis

System Overview Diagram



Early Detection & Treatment Monitoring via Sequential Image Analysis

Progress Overview

Severity Level Classifier Report:

	precision	recall	f1-score	support
mild	0.98	0.98	0.98	447
moderate	0.97	0.96	0.97	447
severe	0.98	0.99	0.98	460
accuracy			0.98	1354
macro avg	0.98	0.98	0.98	1354
weighted avg	0.98	0.98	0.98	1354

Early Detection & Treatment Monitoring via Sequential Image Analysis

Progress Overview

Early Detection & Treatment Monitoring via Sequential Image Analysis

Progress Overview

Early Detection & Treatment Monitoring via Sequential Image Analysis

Requirements

Functional Requirements	Non-Functional Requirements
Capture sequential skin images (Day 1, Day 3, Day 7, etc.)	Accuracy $\geq 90\%$ on benchmark datasets
Preprocess inputs (segmentation, normalization, artifact removal)	Latency ≤ 5 sec on mobile devices
Temporal analysis using CNN + LSTM/GRU/Transformer	Robustness under noisy/low-quality images
Classify condition trends $\rightarrow$ <i>Improving / Stable / Worsening</i>	Fairness across diverse skin tones
Generate reports, heatmaps & alerts	Security: End-to-end encryption & HIPAA/GDPR compliance
Integration with hospital EHR & telemedicine platforms	Usability: Simple mobile interface for patients & doctors

Real world problems **Early Detection & Treatment Monitoring via Sequential Image Analysis**

Scenario

- A patient has a chronic skin condition like **eczema, psoriasis, or acne**.
- They take images at home or in clinics, but **current AI systems analyze only a single image**.
- Doctors cannot objectively see how the condition changes over days or weeks.
- Subtle worsening (increased redness, new lesions) or improvement often goes unnoticed until the next consultation.

Challenges

1.Lack of Continuous Monitoring

- 1.Existing AI cannot track changes over time, missing early signs of worsening.

2.Inconsistent Image Quality

- 1.Variations in lighting, camera angle, or resolution make sequential comparisons difficult.

3.Difficulty Detecting Subtle Changes

- 1.Small changes in lesion size, texture, or color are often invisible to both humans and static-image AI.

Impact

1.Delayed Detection of Disease Progression

- 1.Worsening symptoms may go unnoticed, leading to increased discomfort or complications.

2.Limited Patient Awareness

- 1.Patients cannot visually track how their skin condition is changing over time.

Commercialization

Investment & Cost Recovery

Investment / Cost:

- AI engineers for temporal image analysis and model optimization
- Cloud storage for sequential image data
- Computing resources for continuous monitoring
- Secure APIs and data privacy compliance (health data regulations)

Cost Recovery:

- Freemium model with basic monitoring features
- Low-cost subscription plans for advanced tracking and reports
- Leasing or rental options for organizations with limited budgets

Target Market

- **Sri Lankan patients** (urban & rural) requiring continuous skin condition monitoring
- **Public health sector & NGOs** supporting long-term disease monitoring programs
- **Private clinics and hospitals** for follow-up and remote patient tracking
- **Telemedicine providers** offering remote dermatology services

References

- [1] A. Esteva, et al., “Dermatologist-level classification of skin cancer with deep neural networks,” *Nature*, vol. 542, pp. 115–118, 2017.
- [2] N. C. Codella, et al., “Skin lesion analysis toward melanoma detection: A challenge at ISBI 2016,” *IEEE Transactions on Medical Imaging*, vol. 38, no. 2, pp. 252–265, 2019.
- [3] L.-F. Li, et al., “Deep learning in skin disease image recognition: A review,” *IEEE Access*, vol. 8, pp. 20865–20880, 2020.
- [4] P. Tschandl, et al., “Human–computer collaboration for skin cancer recognition,” *Nature Medicine*, vol. 26, pp. 1229–1234, 2020.
- [5] F. Hameed, et al., “Artificial intelligence in dermatology: A review,” *Cureus*, vol. 13, no. 2, e13544, 2021.
- [6] P. N. Srinivasu, et al., “Classification of skin disease using deep learning neural networks with MobileNetV2 and LSTM,” *Sensors*, vol. 21, no. 8, p. 2852, 2021.
- [7] S. Aksoy, et al., “Robust skin disease classification by distilling deep neural network ensemble for the mobile diagnosis of Herpes Zoster,” *IEEE JBHI*, vol. 26, no. 8, pp. 4074–4085, 2022.
- [8] N. Janbi, et al., “Imtidad: Distributed AI services for skin disease diagnosis over cloud, fog and edge,” *IEEE Transactions on Services Computing*, vol. 15, no. 4, pp. 2278–2291, 2022.
- [9] M. A. Hashmani, et al., “An adaptive federated machine learning-based intelligent system for skin disease detection,” *IEEE Access*, vol. 9, pp. 50148–50160, 2021.

Skin Care Assistant

Overall Progress
Healing Well

Skin Care Assistant
Get Personalized Advice

Skin Health Score

W1W1W3W3W5W5

Symptom Distribution

New Scan

HomeAnalyzeChat

Healing Analysis

Recovery Trend

Good

Fair

Bad

W1W2W3W4W5W6

BadFairGoodTap dot for details

Insight

Your skin condition shows: Positive improvement trend

• Consistency improved across the last weeks

Tip: Continue moisturizer twice daily + avoid harsh soaps for better results.

New Scan

HomeAnalyzeChat

Weekly Progress

Healing Trend

95

80

60

40

30

W1W1W2W2W3W3W4W4W5W5W6W6W7

History

Week 7

Score: 91%

2026-01-04

Improving

Week 6

Score: 95%

2025-12-06

Excellent

Week 5

Score: 85%

2025-11-29

Good

Add Weekly Check-in



It22643322

Wijesinghe H.W.M.O.P

Information Technology

Introduction

- **High prevalence of skin diseases in Sri Lanka**

Common conditions like acne, eczema, and fungal infections affect many people.

- **Limited access to dermatologists**

Rural areas often rely on general practitioners or self-medication, causing misdiagnosis and delayed treatment.

- **Early diagnosis is essential**

Current AI tools rely only on images, missing critical symptoms like itching, pain, burning, or swelling.

- **Multimodal input for better accuracy**

The system accepts **skin images** and **audio-based symptom descriptions** from users for comprehensive analysis.

- **Disease suggestion output**

After processing both visual and audio inputs, the system **classifies symptoms and suggests possible skin diseases**, supporting early and reliable diagnosis.



Problem

1. Visual similarity of skin diseases

1. Image-only classifiers (CNNs) struggle to differentiate diseases that look alike, e.g., acne vs. rosacea.

2. Text-based symptom input limitations

1. Systems requiring typing or literacy exclude rural patients who may have low literacy or are unfamiliar with typing.

3. Dataset bias

1. Most datasets are based on Western populations, leading to poor performance on Sri Lankan skin tones and culturally-specific symptom reporting.

Symptom-Aware Disease Classification



Proposed Solutions

1. Multimodal AI framework

1. Combines **CNN-based image analysis** for skin images and **RNN/speech analysis** for audio symptom descriptions.

2. Fusion layer for integrated analysis

1. Visual and symptom data are combined to improve diagnostic accuracy.

3. Natural speech input in local languages

1. Patients can speak symptoms in **Sinhala or English**, removing literacy and typing barriers.

OBJECTIVES

Main objective:

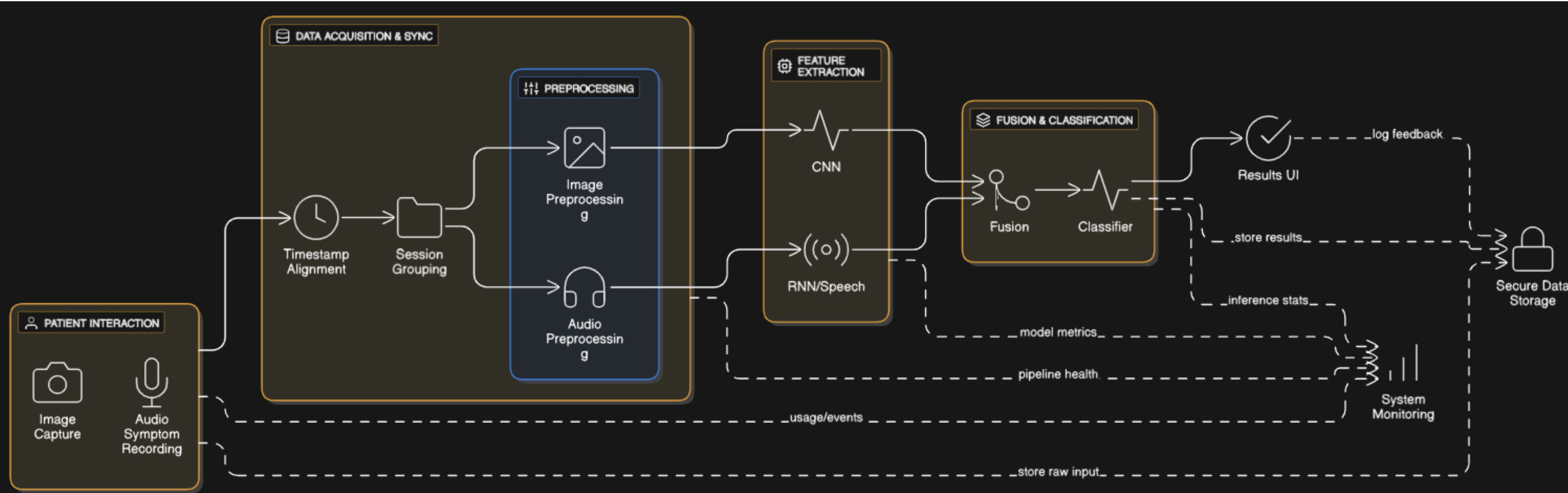
Develop a multimodal AI system for accurate skin disease diagnosis in Sri Lanka by integrating image-based and audio-based symptom inputs to support early and reliable detection, especially for underserved rural populations.



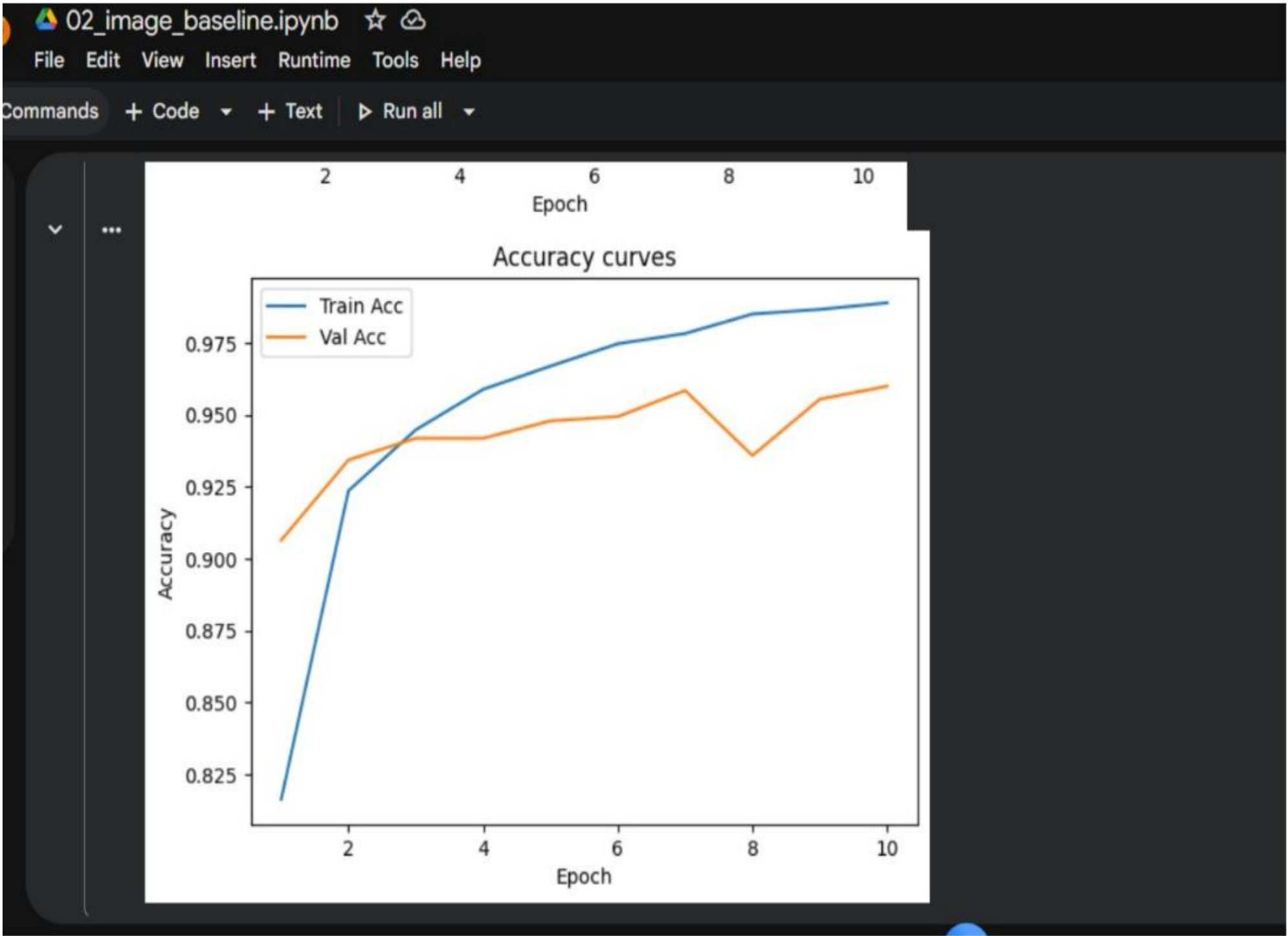
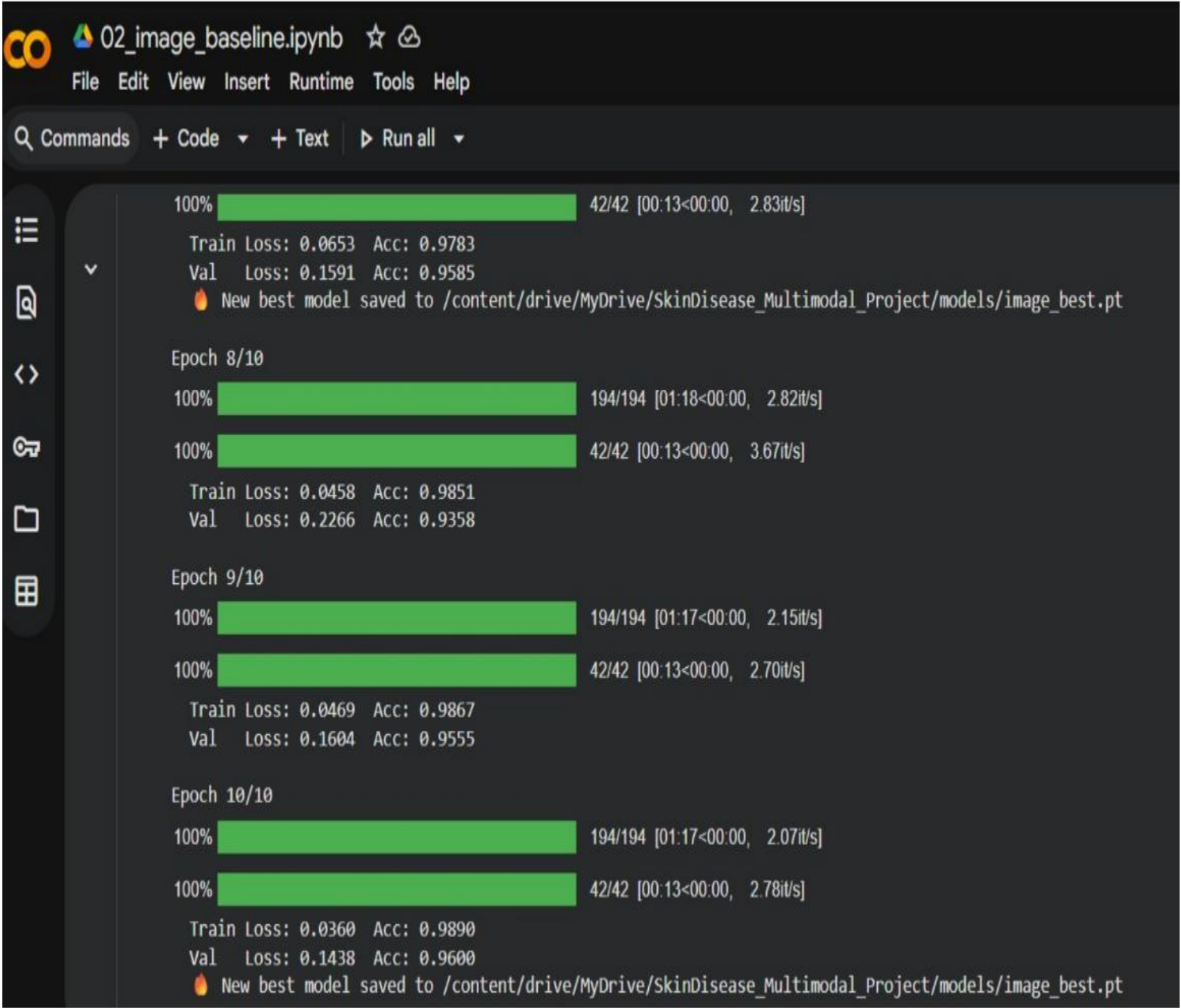
Sub Objectives:

- **Image Analysis** – Classify skin conditions with CNN; ensure accuracy on Sri Lankan skin tones.
- **Audio Symptom Analysis** – Process spoken symptoms (Sinhala/English) with RNN; detect itching, pain, burning, swelling.
- **Multimodal Fusion** – Combine image and audio data for better classification; resolve conflicts between inputs.
- **Accessibility & Usability** – Support natural speech; easy to use for rural patients.
- **Evaluation & Validation** – Test on diverse Sri Lankan datasets; outperform image-only/text-only models.

Technologies Use in the system



Progress Overview



Progress Overview

System Diagram

02b_image_finetune_fixset_PERFECT.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
[ ]
opt.zero_grad()
logits = model(xb)
loss = criterion(logits, yb)
loss.backward()
opt.step()

loss_sum += float(loss.item()) * len(yb)
pred = logits.argmax(1)
correct += int((pred == yb).sum().item())
total += len(yb)

print(f" epoch {ep+1}: loss={loss_sum/total:.4f} acc={correct/total:.4f}")

torch.save(model.state_dict(), CKPT_OUT)
print(" Saved finetuned checkpoint:", CKPT_OUT)
```

epoch 1/5: 100% 250/250 [10:15<00:00, 2.46s/it]
epoch 1: loss=0.6742 acc=0.7362
epoch 2/5: 100% 250/250 [00:46<00:00, 5.37it/s]
epoch 2: loss=0.4386 acc=0.8270
epoch 3/5: 100% 250/250 [00:45<00:00, 5.55it/s]
epoch 3: loss=0.3942 acc=0.8511
epoch 4/5: 100% 250/250 [00:44<00:00, 5.60it/s]
epoch 4: loss=0.3019 acc=0.8829
epoch 5/5: 100% 250/250 [00:46<00:00, 5.42it/s] epoch 5: loss=0.2831 acc=0.8939
Saved finetuned checkpoint: /content/drive/MyDrive/SkinDisease_Multimodal_Project/models/image_best_finetuned.pt

03_audio_baseline.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
[ ]

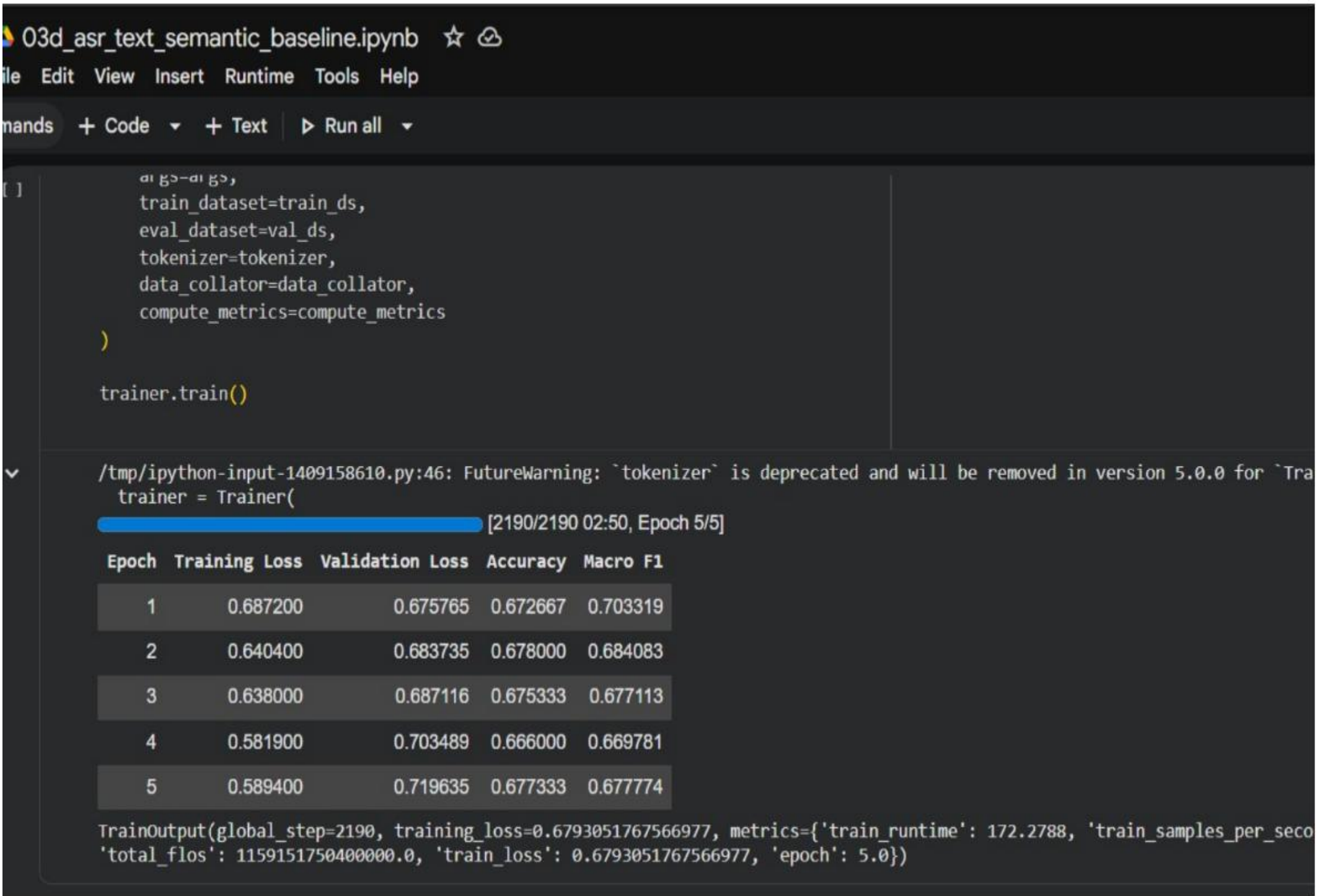
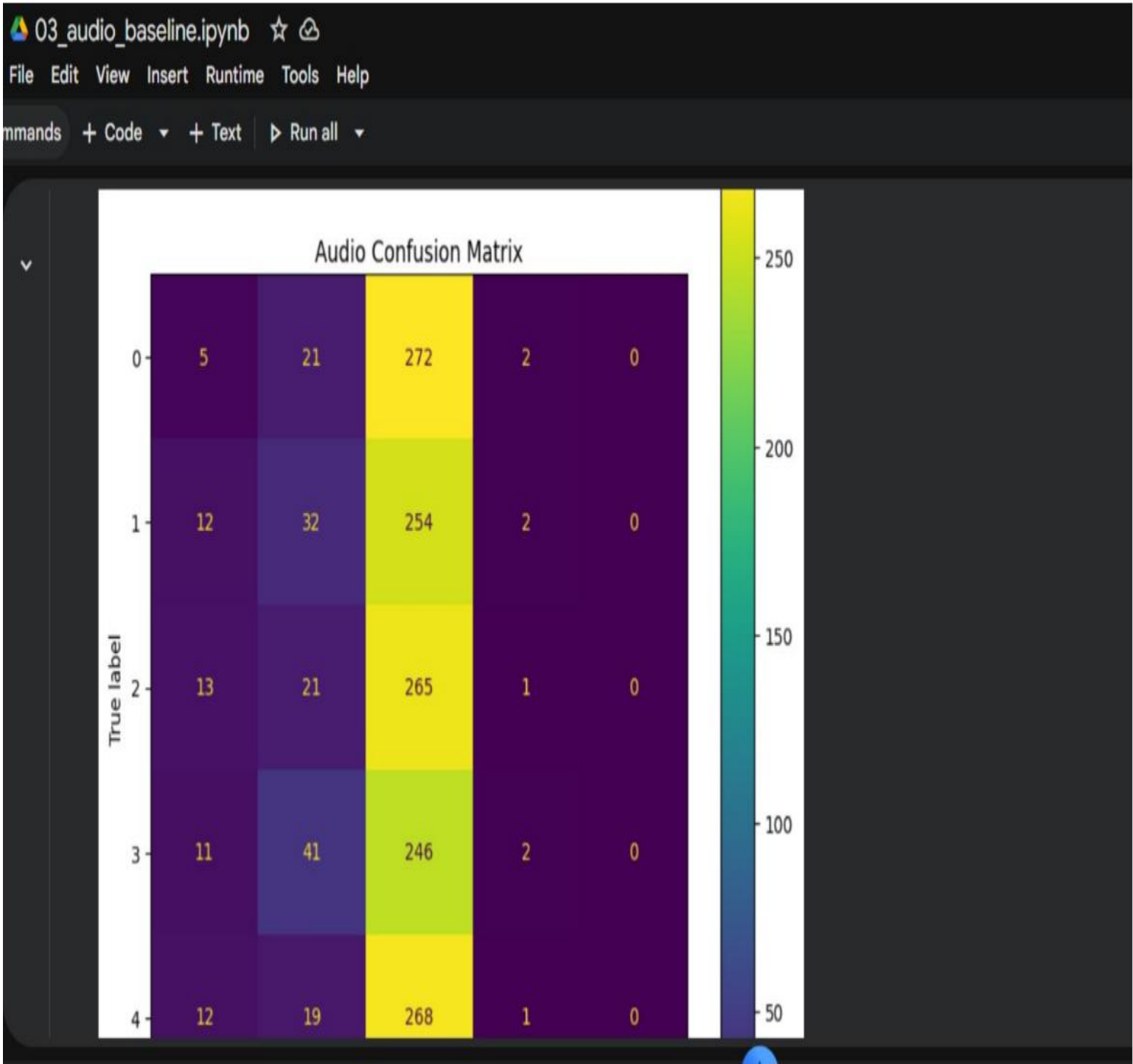
history["train_loss"].append(tr_loss)
history["train_acc"].append(tr_acc)
history["val_loss"].append(va_loss)
history["val_acc"].append(va_acc)

print(f"Epoch {epoch}/{EPOCHS} | Train loss={tr_loss:.4f} acc={tr_acc:.4f} | Val loss={va_loss:.4f} acc={va_acc:.4f}")

if va_acc > best_val_acc:
    best_val_acc = va_acc
    torch.save(model.state_dict(), best_path)
    print(" Saved best model:", best_path)
```

Epoch 1/10 | Train loss=1.6149 acc=0.2093 | Val loss=1.6254 acc=0.1900
Saved best model: /content/drive/MyDrive/SkinDisease_Multimodal_Project/models/audio_best.pt
Epoch 2/10 | Train loss=1.6120 acc=0.2123 | Val loss=1.6192 acc=0.2020
Saved best model: /content/drive/MyDrive/SkinDisease_Multimodal_Project/models/audio_best.pt
Epoch 3/10 | Train loss=1.6093 acc=0.2150 | Val loss=1.6229 acc=0.1867
Epoch 4/10 | Train loss=1.6084 acc=0.2126 | Val loss=1.6244 acc=0.1753
Epoch 5/10 | Train loss=1.6092 acc=0.2183 | Val loss=1.6223 acc=0.1827
Epoch 6/10 | Train loss=1.6086 acc=0.2111 | Val loss=1.6223 acc=0.1833
Epoch 7/10 | Train loss=1.6080 acc=0.2210 | Val loss=1.6195 acc=0.1867
Epoch 8/10 | Train loss=1.6072 acc=0.2167 | Val loss=1.6264 acc=0.1967
Epoch 9/10 | Train loss=1.6083 acc=0.2193 | Val loss=1.6197 acc=0.1913
Epoch 10/10 | Train loss=1.6080 acc=0.2172 | Val loss=1.6222 acc=0.1792

Progress Overview



Functional Requirements

- Capture image + audio inputs.
- Image analysis (CNN).
- Audio analysis (RNN/Transformer).
- Fusion of multimodal features.
- Classify skin/face diseases.
- Provide results + confidence score.
- Clinician dashboard & secure storage.

Non-Functional Requirements

- Accuracy $\geq 90\%$.
- Response time ≤ 10 sec.
- Usability: Simple UI, Sinhala/Tamil voice input.
- Security: Data encryption, HIPAA/GDPR compliance.
- Scalability: Handle many users.
- Cultural adaptability & interpretability.

Software Requirements

- Languages: Python, JavaScript.
- Frameworks: TensorFlow/PyTorch, Librosa, OpenCV.
- Database: MySQL/MongoDB.
- Platforms: Web + Mobile.
- Cloud: AWS/GCP/Azure.

Personnel Requirements

- AI/ML Engineer – Image + audio model development.
- Speech Processing Specialist – Audio feature extraction.
- Dermatologist – Clinical validation & dataset labeling.
- Software Engineer – App & backend development.
- Project Manager – Coordination & integration.

Scenario:

In rural Sri Lanka, a young adult develops **facial redness, bumps, and itching**.

The nearest dermatologist is **over 50 km away**, and local clinics have only general practitioners.

The patient tries **self-medication**, which leads to **worsening of the condition**.

Challenges:

Visual similarity of skin conditions – Acne, rosacea, or fungal infections may look alike; image-only AI may misdiagnose.

Limited literacy or typing ability – Text-based symptom input is not practical for rural patients.

Cultural and skin tone differences – Most datasets are from Western populations, reducing AI accuracy on Sri Lankan skin.

Impact:

Delayed diagnosis and improper treatment.

Increased **pain, discomfort, and risk of complications**.

Healthcare inequality between urban and rural areas.

Solution via Your System:

Patient captures a **photo of the affected area** and **speaks symptoms naturally** in Sinhala/English.

The multimodal AI **analyzes image + audio** and **suggests possible skin diseases**.

Enables **early, accurate, and accessible diagnosis**, even in remote areas.

- **Investment & Cost Recovery**

- Initial: Development team, dermatology experts, cloud servers.
- Recovery: SaaS model for clinics + mobile app subscriptions for patients.

- **Target Market**

- Hospitals & clinics (decision support).
- Dermatologists (efficiency & accuracy).
- Patients (affordable diagnosis support, especially rural).

- **Value Proposition**

- First **symptom-aware multimodal dermatology AI** in Sri Lanka.
- Supports natural audio input → inclusive for low-literacy users.
- Improves accuracy by integrating **images + real patient narratives**.

- **IP Rights**

- Patent on multimodal audio+image fusion method.
- Trademark protection for system brand/app.

- [1] A. Yang and E. Yang, “A Multimodal Approach to the Detection and Classification of Skin Diseases,” *arXiv*, Nov. 2024.
- [2] S. Hamida et al., “An Integrated Multimodal Deep Learning System for Dermatology Diagnosis,” *IEEE Access*, vol. 12, pp. 4521–4535, 2024.
- [3] X. Zhang et al., “Mobile AI Dermatology Applications: Performance and Limitations,” *JMIR Dermatology*, 2023.
- [4] R. Gupta et al., “Multimodal Fusion for Clinical Decision Support: Combining Images and Patient Data,” *Nature Digital Medicine*, 2024.
- [5] A. Esteva, B. Kuprel, R.A. Novoa, J. Ko, S.M. Swetter, H.M. Blau, and S. Thrun, “Dermatologist-level Classification of Skin Cancer with Deep Neural Networks,” *Nature*, vol. 542, pp. 115–118, 2017.

←

New Scan

Upload a clear photo of the affected area.

Tap to select image

Camera

Gallery

Describe your symptoms (Voice)

Send & Analyze

←

Analysis Result

Eczema

Confidence: 85.0%

Common Symptoms

Redness

Itching

Dryness

Recommended Routine

Morning

Gentle Cleanser, Moisturizer

Night

Topical Corticosteroid (if prescribed), Heavy Cream

Treatment

Use prescribed ointment twice daily

If bleeding occurs, see a doctor immediately.

IT22249784
DEVINDI K.T.P

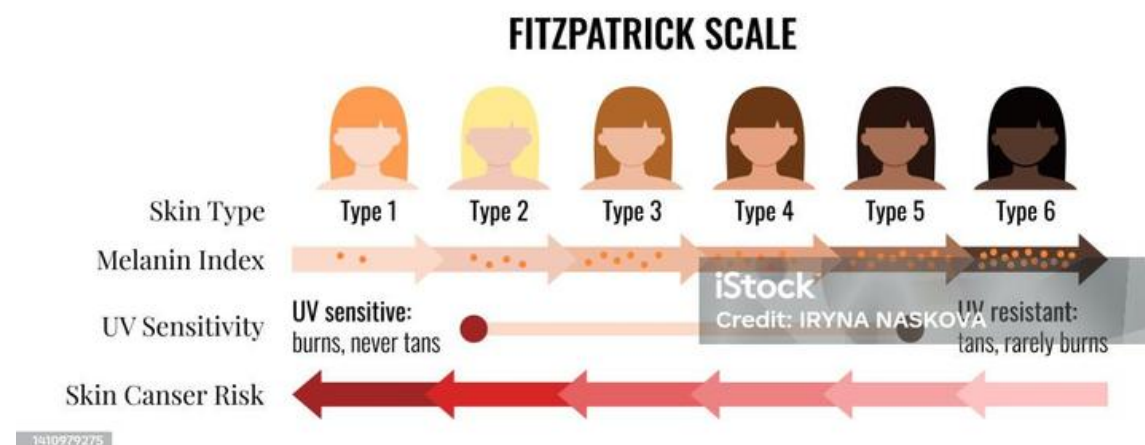


B.Sc. (Hons) Degree in Information Technology
Specializing in Information Technology

Cross-Cultural Disease Classification via Domain Adaptation

INTRODUCTION

- Addresses **bias in skin disease AI** across light and dark skin tones
- Applies **domain adaptation** to ensure fair and accurate classification
- Includes **skin type classification** for melanin-aware diagnosis
- Integrates a **personalized skincare assistant** for individual needs
- Supports **culturally-aware healthcare** in Sri Lanka and diverse populations



Problem & Solution:

Problem

Existing skin disease AI models are **biased toward light skin tones** due to unbalanced training datasets

Poor performance on **dark and diverse skin types** leads to inaccurate or unreliable diagnoses

Lack of **skin type awareness** reduces model generalization across ethnic groups

Current systems rarely provide **personalized skincare guidance** after diagnosis

Limited support for **culturally-aware healthcare**, especially in regions like Sri Lanka

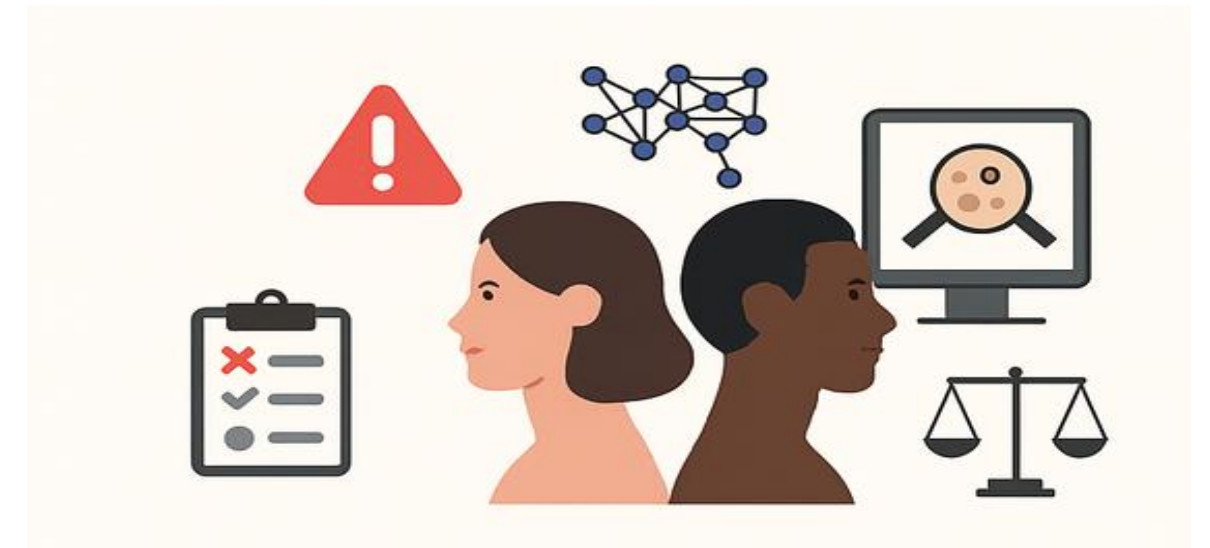
Solution

Apply **domain adaptation techniques** to reduce bias between different skin tone datasets

Integrate **skin type classification** to enable melanin-aware disease detection

Train and evaluate models using **diverse, cross-cultural datasets**

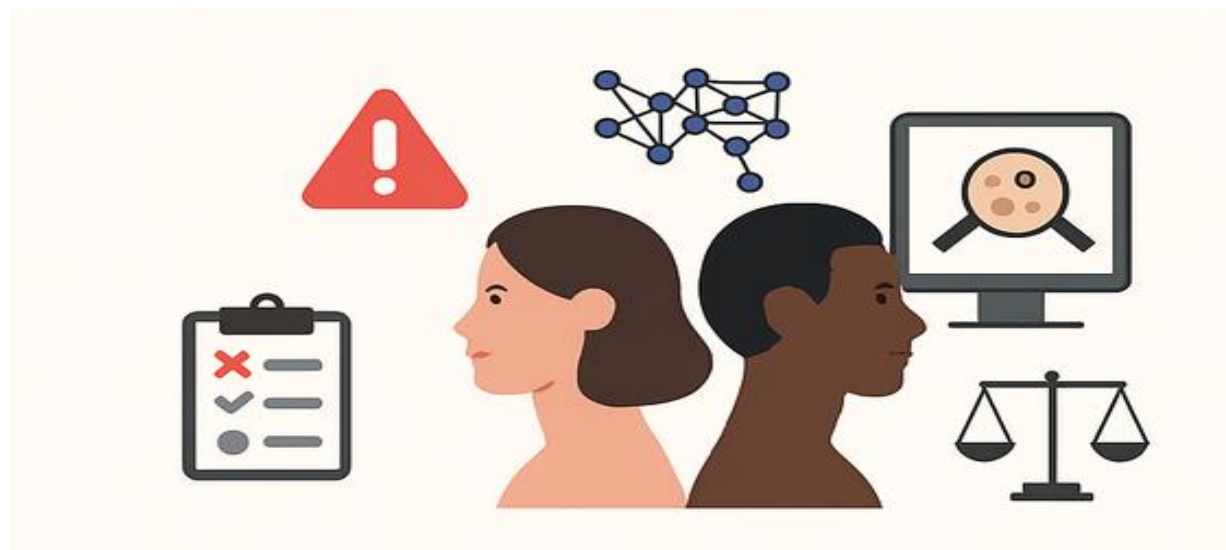
Introduce a **personal skincare assistant** tailored to individual skin type and condition



OBJECTIVES

Main Objectives

To develop a **fair and inclusive skin disease classification system** using **domain adaptation**, ensuring accurate diagnosis across diverse skin tones and cultural populations.



Sub Objectives

- Reduce **model bias** between light and dark skin tone datasets through domain adaptation techniques
- Perform **skin type classification** to enable melanin-aware disease prediction
- Improve classification accuracy for **underrepresented ethnic groups**, especially South Asian skin types
- Develop a **personalized skincare assistant** based on detected skin type and condition
- Support **culturally-aware and accessible healthcare** for Sri Lanka and global communities



Technologies Used

AI / Machine Learning-

PyTorch , timm (PyTorch Image Models) torchvision, Pillow

Model Architectures-

ConvNeXt (Final selected model) MobileNetV3 Vision Transformer (ViT) Swin Transformer

Backend/Frontend

Flutter ,FastAPI ,Uvicorn

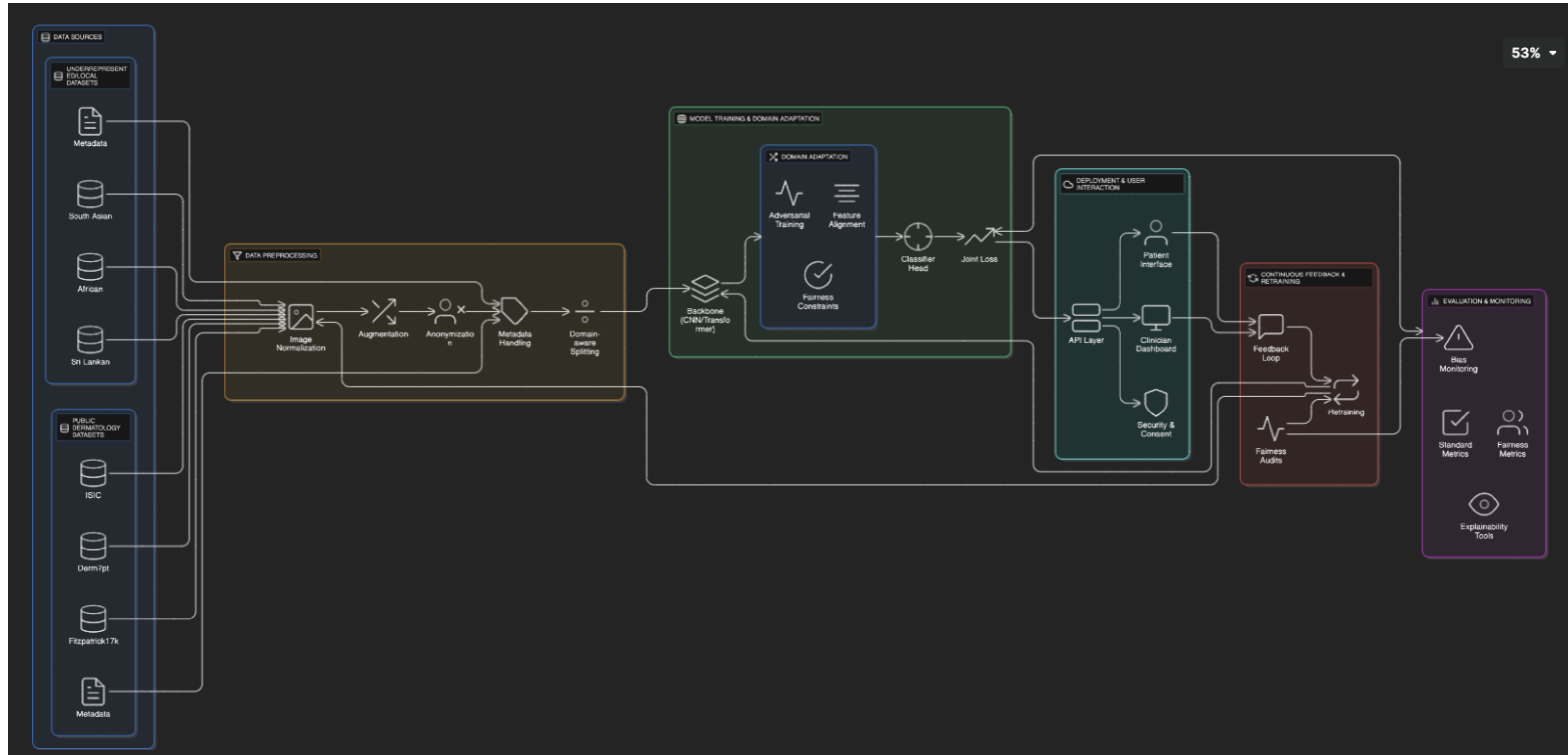
Version Control & Collaboration

Git



Implementation of the project

• SYSTEM OVERVIEW DIAGRAM



Progress Overview

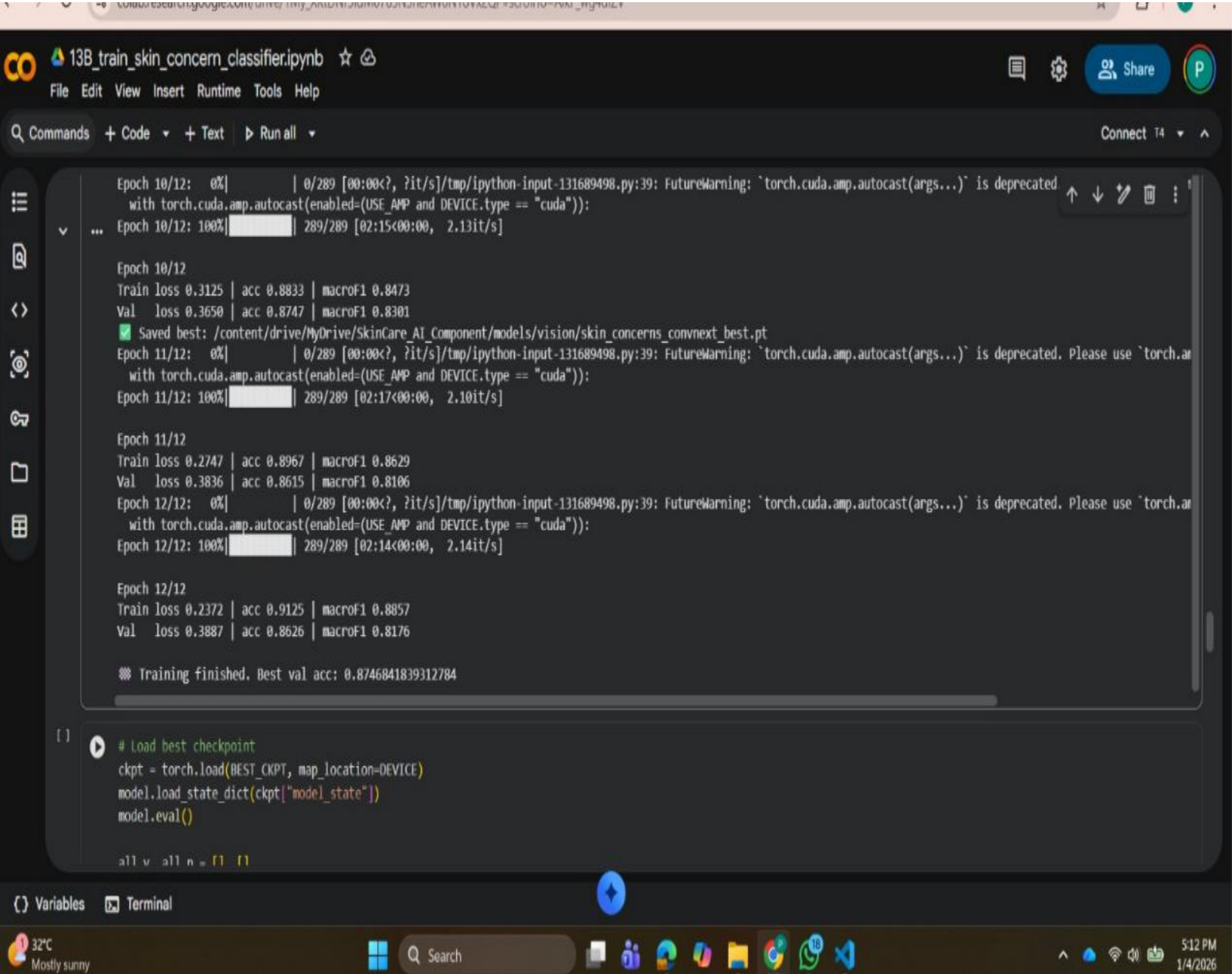
The screenshot shows a Google Colab notebook titled "13C_final_personal_skin_care_assistant_demo.ipynb". The code defines a function to handle image input, either from a laptop or a webcam. The output shows the user selecting "1" for "Upload from laptop", and a file named "WhatsApp Image 2025-12-24 at 20.05.00.jpeg" is uploaded. The notebook also displays a small image of a person's face, labeled "Input Image".

The screenshot shows a Google Colab notebook titled "12_train_skin_type_convnext_cleaned.ipynb". The code displays the training progress of a ConvNeXt model for skin type classification. The training log shows the following results:

Epoch	train acc	val acc
10/20	0.5857	0.5448
11/20	0.6456	0.6279
12/20	0.6996	0.5763
13/20	0.7431	0.6718
14/20	0.8085	0.6870
15/20	0.8410	0.7109
16/20	0.8778	0.7023
17/20	0.8935	0.7233
18/20	0.9076	0.7166
19/20	0.9260	0.6641
20/20	0.9197	0.7405

The code also shows the best validation accuracy achieved: 0.7404580152671756. The final code block shows the model being loaded and evaluated on a test set.

Progress Overview



```
Epoch 10/12: 0% | 0/289 [00:00<?, ?it/s]/tmp/ipython-input-131689498.py:39: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated
with torch.cuda.amp.autocast(enabled=(USE_AMP and DEVICE.type == "cuda")):
Epoch 10/12: 100% | 289/289 [02:15<00:00, 2.13it/s]

Epoch 10/12
Train loss 0.3125 | acc 0.8833 | macroF1 0.8473
Val loss 0.3650 | acc 0.8747 | macroF1 0.8301
✓ Saved best: /content/drive/MyDrive/SkinCare_AI_Component/models/vision/skin_concerns_convnext_best.pt
Epoch 11/12: 0% | 0/289 [00:00<?, ?it/s]/tmp/ipython-input-131689498.py:39: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp
with torch.cuda.amp.autocast(enabled=(USE_AMP and DEVICE.type == "cuda")):
Epoch 11/12: 100% | 289/289 [02:17<00:00, 2.10it/s]

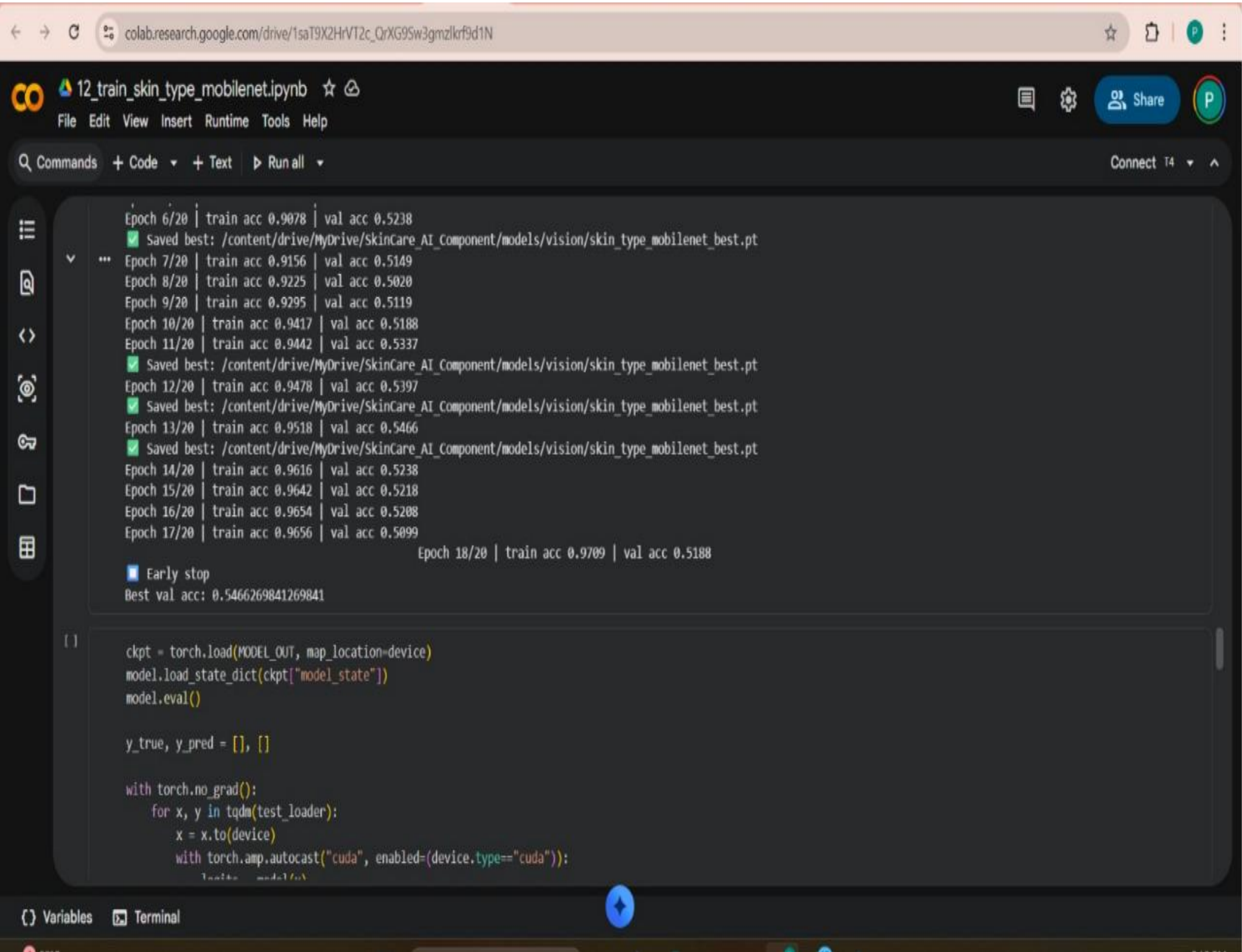
Epoch 11/12
Train loss 0.2747 | acc 0.8967 | macroF1 0.8629
Val loss 0.3836 | acc 0.8615 | macroF1 0.8106
Epoch 12/12: 0% | 0/289 [00:00<?, ?it/s]/tmp/ipython-input-131689498.py:39: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp
with torch.cuda.amp.autocast(enabled=(USE_AMP and DEVICE.type == "cuda")):
Epoch 12/12: 100% | 289/289 [02:14<00:00, 2.14it/s]

Epoch 12/12
Train loss 0.2372 | acc 0.9125 | macroF1 0.8857
Val loss 0.3887 | acc 0.8626 | macroF1 0.8176

☛ Training finished. Best val acc: 0.8746841839312784

[1] # Load best checkpoint
ckpt = torch.load(BEST_CHKPT, map_location=DEVICE)
model.load_state_dict(ckpt["model_state"])
model.eval()

all v all n = [1] [1]
```



```
Epoch 6/20 | train acc 0.9078 | val acc 0.5238
✓ Saved best: /content/drive/MyDrive/SkinCare_AI_Component/models/vision/skin_type_mobilenet_best.pt
Epoch 7/20 | train acc 0.9156 | val acc 0.5149
Epoch 8/20 | train acc 0.9225 | val acc 0.5020
Epoch 9/20 | train acc 0.9295 | val acc 0.5119
Epoch 10/20 | train acc 0.9417 | val acc 0.5188
Epoch 11/20 | train acc 0.9442 | val acc 0.5337
✓ Saved best: /content/drive/MyDrive/SkinCare_AI_Component/models/vision/skin_type_mobilenet_best.pt
Epoch 12/20 | train acc 0.9478 | val acc 0.5397
✓ Saved best: /content/drive/MyDrive/SkinCare_AI_Component/models/vision/skin_type_mobilenet_best.pt
Epoch 13/20 | train acc 0.9518 | val acc 0.5466
✓ Saved best: /content/drive/MyDrive/SkinCare_AI_Component/models/vision/skin_type_mobilenet_best.pt
Epoch 14/20 | train acc 0.9616 | val acc 0.5238
Epoch 15/20 | train acc 0.9642 | val acc 0.5218
Epoch 16/20 | train acc 0.9654 | val acc 0.5208
Epoch 17/20 | train acc 0.9656 | val acc 0.5099
Epoch 18/20 | train acc 0.9709 | val acc 0.5188

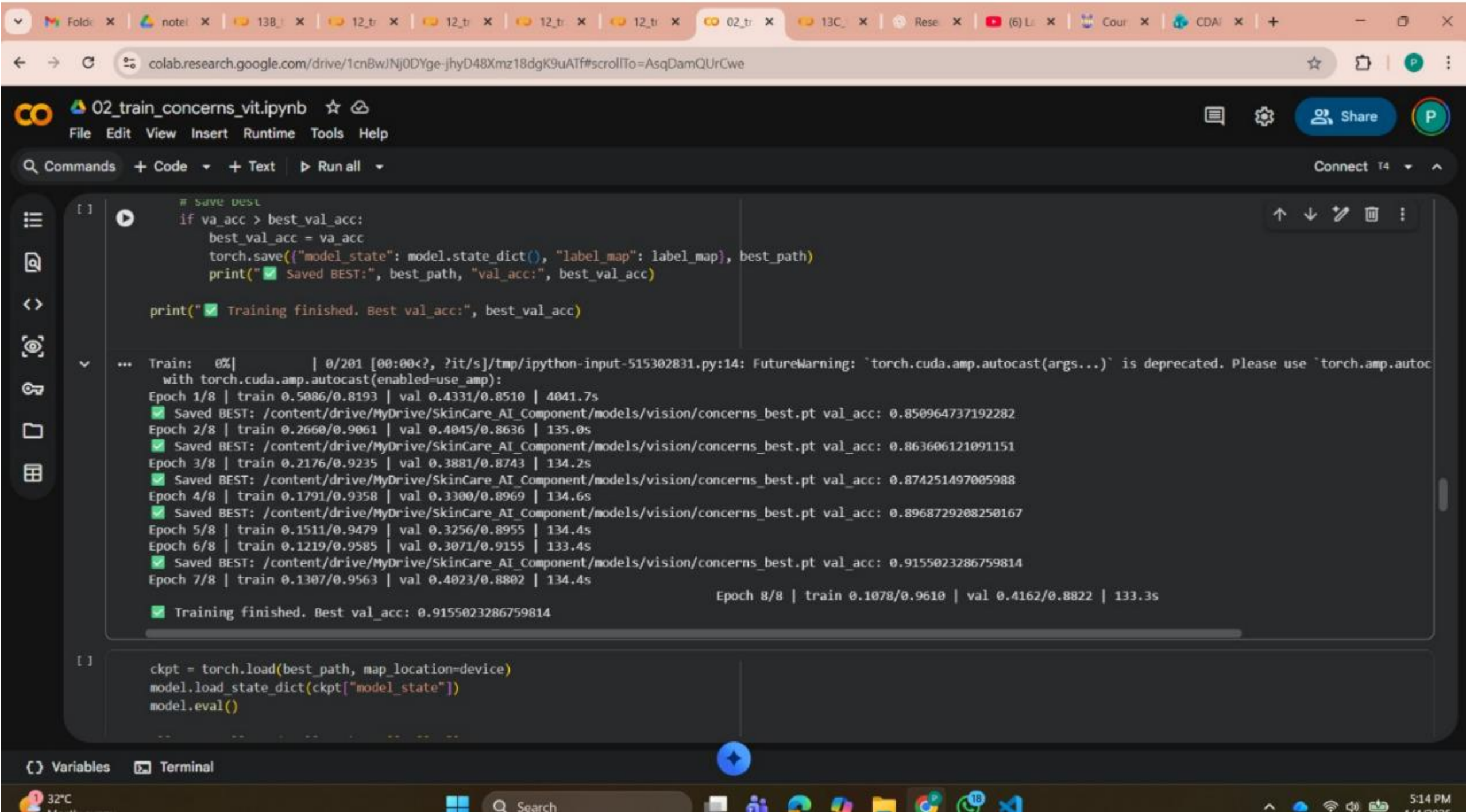
☑ Early stop
Best val acc: 0.5466269841269841

[1] ckpt = torch.load(MODEL_OUT, map_location=device)
model.load_state_dict(ckpt["model_state"])
model.eval()

y_true, y_pred = [], []

with torch.no_grad():
    for x, y in tqdm(test_loader):
        x = x.to(device)
        with torch.amp.autocast("cuda", enabled=(device.type=="cuda")):
            y_hat = model(x)
```

Progress Overview



```
# Save BEST
if va_acc > best_val_acc:
    best_val_acc = va_acc
    torch.save({"model_state": model.state_dict(), "label_map": label_map}, best_path)
    print("✅ Saved BEST:", best_path, "val_acc:", best_val_acc)

print("✅ Training finished. Best val_acc:", best_val_acc)
```

```
... Train: 0% | 0/201 [00:00<?, ?it/s] /tmp/ipython-input-515302831.py:14: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast` with `torch.cuda.amp.autocast(enabled=use_amp)`:
Epoch 1/8 | train 0.5086/0.8193 | val 0.4331/0.8510 | 4041.7s
✅ Saved BEST: /content/drive/MyDrive/SkinCare_AI_Component/models/vision/concerns_best.pt val_acc: 0.850964737192282
Epoch 2/8 | train 0.2660/0.9061 | val 0.4045/0.8636 | 135.0s
✅ Saved BEST: /content/drive/MyDrive/SkinCare_AI_Component/models/vision/concerns_best.pt val_acc: 0.863606121091151
Epoch 3/8 | train 0.2176/0.9235 | val 0.3881/0.8743 | 134.2s
✅ Saved BEST: /content/drive/MyDrive/SkinCare_AI_Component/models/vision/concerns_best.pt val_acc: 0.874251497005988
Epoch 4/8 | train 0.1791/0.9358 | val 0.3300/0.8969 | 134.6s
✅ Saved BEST: /content/drive/MyDrive/SkinCare_AI_Component/models/vision/concerns_best.pt val_acc: 0.8968729208250167
Epoch 5/8 | train 0.1511/0.9479 | val 0.3256/0.8955 | 134.4s
Epoch 6/8 | train 0.1219/0.9585 | val 0.3071/0.9155 | 133.4s
✅ Saved BEST: /content/drive/MyDrive/SkinCare_AI_Component/models/vision/concerns_best.pt val_acc: 0.9155023286759814
Epoch 7/8 | train 0.1307/0.9563 | val 0.4023/0.8802 | 134.4s
Epoch 8/8 | train 0.1078/0.9610 | val 0.4162/0.8822 | 133.3s
✅ Training finished. Best val_acc: 0.9155023286759814
```

```
ckpt = torch.load(best_path, map_location=device)
model.load_state_dict(ckpt["model_state"])
model.eval()
```

Functional Requirements

- Accept skin images from patients.
- Classify disease fairly across all skin tones.
- Provide explainable output (e.g., Grad-CAM).
- Continuous feedback loop to retrain and reduce bias.

Non-Functional Requirements

- Accuracy & Fairness: >85% accuracy, balanced error across tones.
- Performance: Fast (<3s) + reliable (uptime >99%).
- Security: Data encryption, consent handling.
- Cultural Adaptation: Clear Sinhala/English output, Sri Lanka-specific.

Software Requirements

- Backend: Python (FastAPI) for APIs.
- Frontend: Web interface (React).
- Database: PostgreSQL/MongoDB for patient + training records.
- AI Models: CNNs + Domain Adaptation (DANN, ADDA, FairDisCo).
- Explainability: Grad-CAM / SHAP.
- Security: Encryption, access control, audit logs.

Human Resource Requirements

- AI/ML Engineer: Build & fine-tune domain adaptation models.
- Backend Developer: Implement APIs + database.
- Frontend Developer: Patient-facing UI.
- Database Admin (DBA): Secure, reliable record storage.

Real World Problem

Scenario

A patient with a **dark skin tone in Sri Lanka** develops eczema.

Existing AI dermatology tools, trained mostly on light-skinned datasets, **misclassify the condition**.

Without accurate AI assistance, the patient **experiences delayed treatment**, discomfort, or worsening symptoms.

Challenge

Most AI skin disease models are trained on datasets dominated by **light-skinned individuals**.

Models **struggle to generalize** to darker skin tones due to **feature differences** (e.g., color contrast, lesion visibility).

Limited **ethnic and regional data** (e.g., South Asian, Sri Lankan skin types) hinders model reliability.

Incorporating **skin type variations** and **cultural skincare practices** is complex for AI systems.

Impact

Misdiagnosis or delayed diagnosis for people with darker skin tones.

Reduced **trust in AI healthcare solutions**, potentially widening healthcare inequality.

Patients may **receive incorrect treatment or skincare advice**, affecting outcomes.

Overall, healthcare systems fail to deliver **fair and inclusive dermatological support**.

Commercialization

Investment & Cost Recovery

- **Investment/Cost:**
 - Investment/Cost: AI engineers, cloud hosting, APIs, and compliance.
 - Cost Recovery: Freemium model + low-cost subscriptions.

Target Market:

- Sri Lankan patients (urban & rural).
- Public health sector & NGOs.
- Private clinics, pharmacies, insurers.



Customer Profile:

- Patients needing affordable diagnosis.
- Parents managing children's skin issues.
- Healthcare providers needing fair triage support.

Value Proposition:

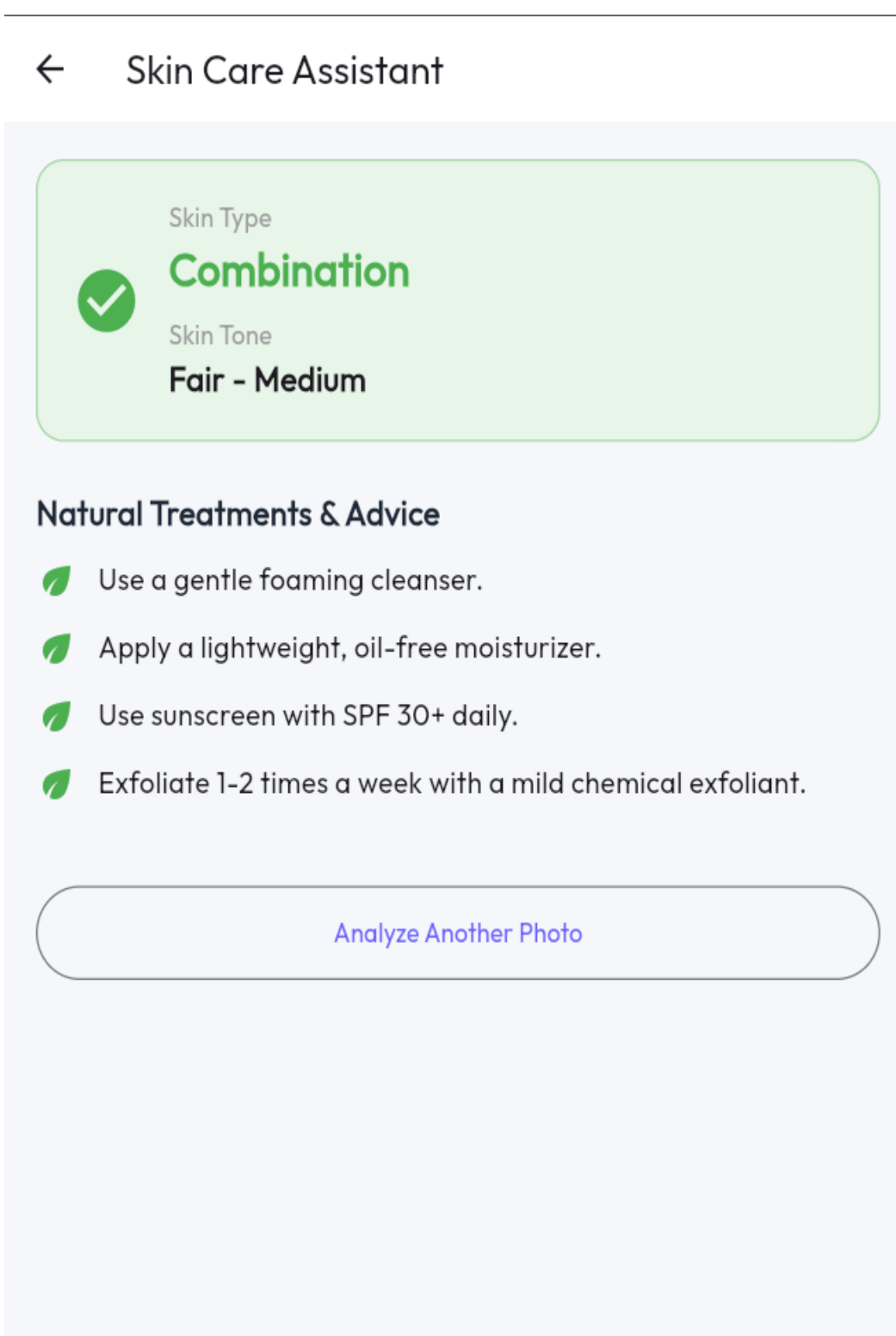
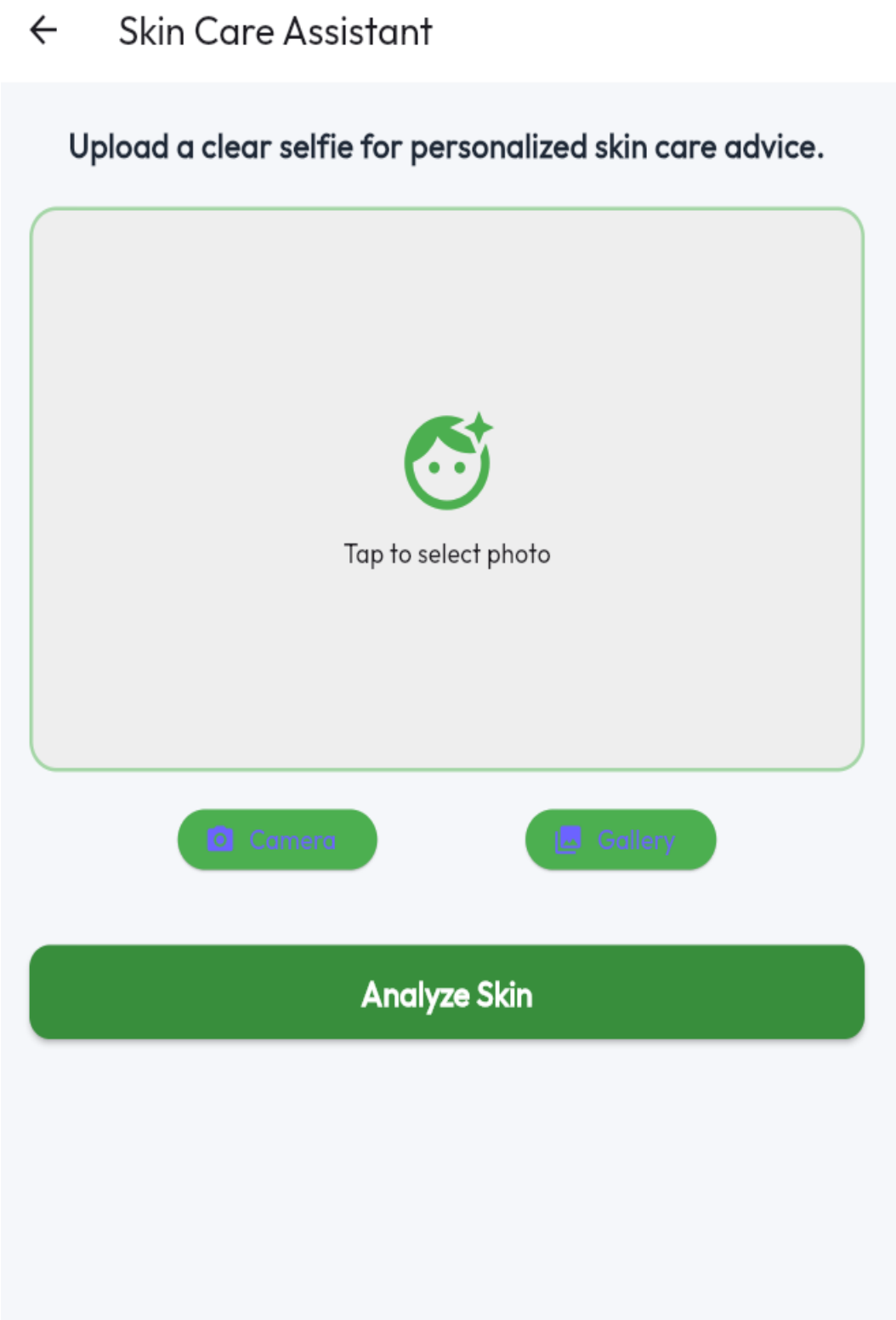
- Fair, accessible, and inclusive AI diagnosis.
- Multilingual and culturally adapted for Sri Lanka.
- Reduces misdiagnosis risk and unnecessary visits.

Intellectual Property Rights (IPR):

- Trademark system name.
- Copyright datasets and UI.
- Patent for fairness-driven domain adaptation pipeline (if possible).

REFERENCES

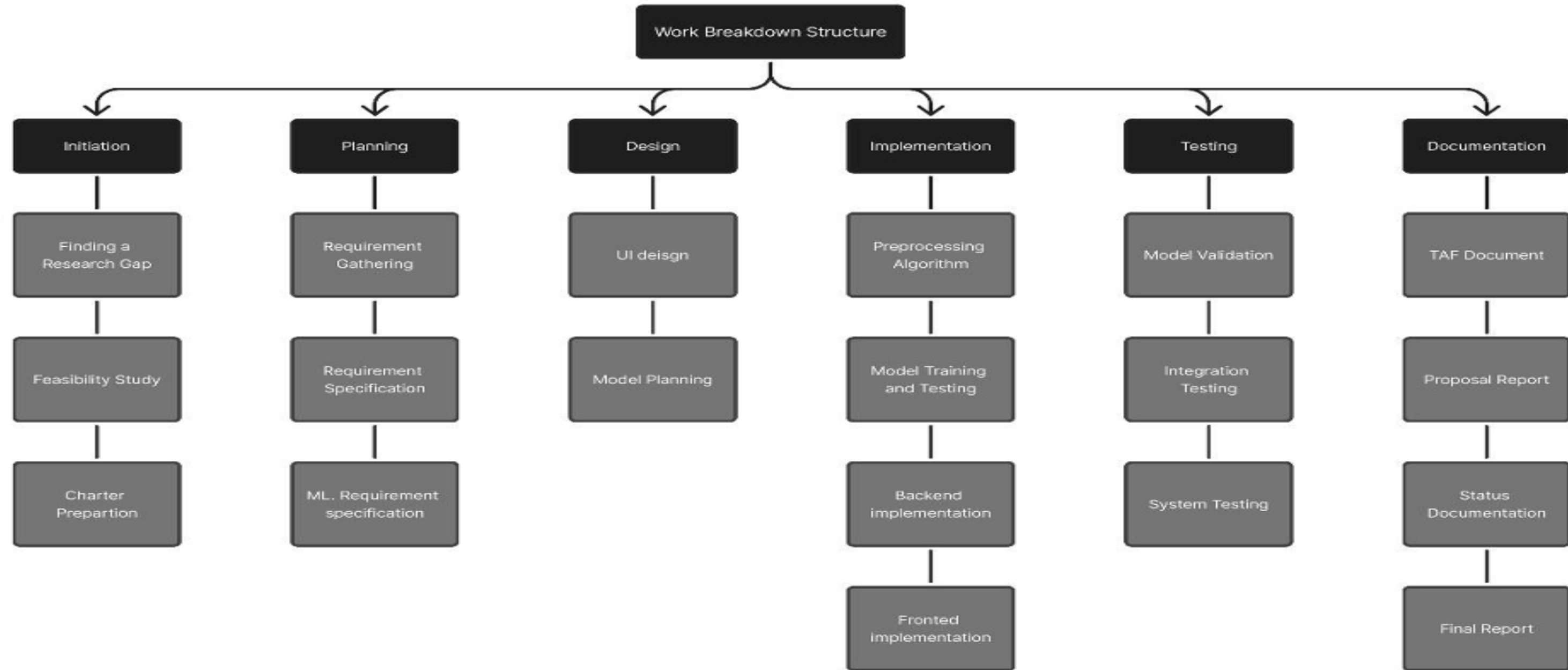
- [1] García-de-la-Puente, N. P., et al. (2024). Domain adaptation for unsupervised cancer detection: An application for skin whole slide images from an interhospital dataset. In MICCAI 2024. [Linkpapers.miccai.org+1](https://linkpapers.miccai.org+1)
- [2] Wang, J., Zhang, Y., Ding, Z., & Hamm, J. (2024). Achieving reliable and fair skin lesion diagnosis via unsupervised domain adaptation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 5157– 5166. [LinkCVF Open Access](#)
- [3] Pombo, G., et al. (2023). Equitable modelling of brain imaging by counterfactual augmentation with morphologically constrained 3D deep generative models. *Medical Image Analysis*, 84, 102723.
- [4] Groh, C., et al. (2022). Towards transparency in dermatology image datasets with skin tone annotations by experts, crowds, and an algorithm. *Proceedings of the ACM on Human-Computer Interaction*, 6, 1–25.
- [5] Bevan, P., & Atapour-Abarghouei, A. (2022). Detecting melanoma fairly: Skin tone detection and debiasing for skin lesion classification. In MICCAI Domain Adaptation & Rep. Transfer Workshop.
- [6] Groh, C., et al. (2022). Towards transparency in dermatology image datasets with skin tone annotations by experts, crowds, and an algorithm. *Proceedings of the ACM on Human-Computer Interaction*, 6, 1–25.
- [7] Stanley, M., Wilms, M., & Forkert, N. (2022). Disproportionate subgroup impacts and other challenges of fairness in artificial intelligence for medical image analysis. In *Ethical and Philosophical Issues in Medical Imaging*.



GRANT CHART

ID	Task Name															
		July	Aug	Sep	Oct	Nov	Dec	Jan-2026	Feb-2026	March-2026	April-2026	May-2026	June2026	July-2026	Aug-2026	
1	Topic Finding															
2	Requirement gathering & Analysis															
3	Topic submission & confirmation															
4	Literature Review															
4	Project proposal draft															
5	Proposal presentation															
6	Project proposal report															
7	Progress Presentation 50%															
8	Research Paper															
9	Progress Presentation 90%															
10	Research project website															
11	Final Report															
12	Final presentation & viva															

BUDGET AND BUDGET JUSTIFICATION



BUDGET AND BUDGET JUSTIFICATION

Category	Cost (LKR)
Research Assistants (2 students, part-time, 6 months)	Rs. 1,320,000
Dermatologist consultant (dataset validation)	Rs. 825,000
Cloud computing (GPU servers ~6 months)	Rs. 990,000
Software tools/licenses (APIs, NLP/LLM, audio tools)	Rs. 495,000
Dermatology image datasets (acquisition, annotation)	Rs. 660,000
Audio dataset creation/simulation	Rs. 330,000
Validation with diverse populations	Rs. 495,000
Admin & dissemination (conference, workshops, printing)	Rs. 330,000
Total Estimated Budget	Rs. 5,445,000

Thank You